User's Manual

MALIB ver.1.3.0

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Japan Aerospace Exploration Agency (JAXA)

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1. Overview

MALIB, which is fork program of RTKLIB[1], is an open-source program package especially for MADOCA[2]-PPP[3]. In this document, we explain the differences between MALIB and RTKLIB 2.4.3 b34 and show how the MADOCA-PPP works in MALIB. We hope it will help your better understand. The key features of MALIB are:

(1) Implementation based on the latest version of RTKLIB (2.4.3 b34 revision)
(2) Supports real-time processing and post-processing
(3) Supports of PPP/PPP-AR[4] with multi-GNSS (GPS/GLONASS/Galileo/QZSS)
(4) Supports multiple receiver L6E data formats (Ublox (ubx) / Septentrio (sbf) / MSJ) and L6E archive data published by the Cabinet Office
(5) Allows frequency selection for positioning calculations per GNSS satellite system
(6) Supports PPP/PPP-AR with local area atmospheric corrections via Internet
(7) Enables two alternative positioning techniques when L6E data is unavailable
(8) Implements JAXA MADOCA GNSS attitude model for precise orbit determination
(9) Uses dead reckoning to maintain positioning continuity during GNSS signal outages
(10) Implementation of PPP positioning utilizing INS-based inertial navigation
(11) Implements local correction information interpolation software
(12) Supports L6D wide-area ionospheric correction information
(13) Implements NLOS observation exclusion function

[1] RTKLIB : An Open-Source Program Package for GNSS Positioning, https://www.rtklib.com/
[2] MADOCA : Multi-GNSS Advanced Orbit and Clock Augmentation
[3] PPP : Precise Point Positioning
[4] AR : Ambiguity Resolution

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2. License

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3. Instructions

3.1. Installation

(1) Extract the software package MALIB.<ver>.zip to an appropriate directory <install dir> (<ver> indicates the version number).

The directory structure of MALIB after extraction is as follows:

```
MALIB.<ver>
├── app                 : Build environment for APs
│   └── consapp         : Console Applications
│       ├── lclcmbpp    : Post-processing Local Correction Information Interpolation
│       ├── lclcmbrt    : Real-time Processing Local Correction Information Interpolation
│       ├── recvbias    : Receiver Code Bias Computation
│       ├── rnx2rtkp    : Post processing
│       └── rtkrcv      : Real-time processing
├── bin                 : Executable binaries and sample configuration files
├── doc                 : Documentation (Manual of MALIB / Release Notes)
├── lib                 : Library
│   └── mkl             : Intel optimized math library
└── src                 : Source programs of MALIB library
    └── rcv             : Source programs depending on GNSS receivers
```

(2) Compile the source code using the makefile located at MALIB/app/consapp to build the executable applications.

(3) Recommended System Requirements

To ensure optimal performance, we recommend the following system configuration:

- Operating System : Ubuntu 20.04
- Compiler : gcc 9.4.0
- Libraries : Included in /lib

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3.2. Additional options

The new configuration options added to MALIB are listed in Table 1. In addition, "-rst" shown in Table 3 was added as a command for real-time processing (rtkrcv).

Table 1 Configuration options added in MALIB

No.	Item	Configuration File	Descriptions
1	Receiver Dynamics	pos1-dynamics	Defines the dynamics model of the target receiver. (0:off,1:dynamics,2:ins) -off: Dynamics is not used -dynamics: Receiver velocity and acceleration are estimated. The receiver position is predicted with the estimated velocity and acceleration. -ins: Performs PPP positioning using INS-based inertial navigation.
2	Substitute PPP Option	pos1-subopt	Defines the substitute PPP mode selection. (0:off,1:brdc, 4:rt) -OFF : Not applied -BRDC : Uses broadcast ephemeris -RT : Uses real-time ephemeris
3	Dead Reckoning Interval	pos1-drint	Defines the dead reckoning interval in seconds. If a value greater than 0 is Defines the, this setting will be used.
4	SNR weighting threshold	pos1-snrwthres_gps	Specify the value in the format "XX = threshold." XX represents the signal identifier in RINEX format (numerical value + alphabet). (Table 8) Defines the SNR weighting threshold. Used to determine the validity of satellite signals based on elevation angle. The threshold values correspond to elevation angle intervals of 5° (ex., 0–5°, 10°, 15° … 85–90°) and should be specified as comma-separated values. Configure individually for each satellite system and signal type.
5		pos1-snrwthres_glo	
6		pos1-snrwthres_gal	
7		pos1-snrwthres_qzs	
8		pos1-snrwthres_bds	
9	SNR Weighting Margin	pos1-snrwmargin_gps	Defines the SNR weighting margin. The SNR weighting margin represents the valid range (in dB) for satellite signals whose SNR falls below the SNR weighting threshold. Configure individually for each satellite system.
10		pos1-snrwmargin_glo	
11		pos1-snrwmargin_gal	
12		pos1-snrwmargin_qzs	
13		pos1-snrwmargin_bds	
14	SNR Weighting Error Standard	pos1-snrweratio_gps	Defines the SNR weighting error standard deviation increase rate.

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No.	Item	Configuration File	Descriptions
15	Deviation Increase Rate	pos1-snrweratio_glo	defines the rate at which the observation error standard deviation increases per dB within the range below the SNR weighting threshold (i.e., within the SNR weighting margin). For example: • If the threshold is 45 dB, the margin is 10 dB, and the increase rate is 10, • Then a signal with 44 dB SNR will have an error multiplier of 10 applied.
16		pos1-snrweratio_gal	
17		pos1-snrweratio_qzs	
18		pos1-snrweratio_bds	
19	Wide-Area Ionospheric Correction Information	pos2-ionocorr	Defines whether to use wide-area ionospheric correction information.(0:off,1:on)
20	GNSS Used for PPP-AR	pos2-arsys	Set the satellite systems to be used for integer ambiguity resolution using a bit value. (Bit Array: 1:gps + 8:gal + 16:qzs + 32:bds)
21	Cascading Ambiguity Resolution	pos2-car	CAR (Cascading Ambiguity Resolution) enable flag.
22	CAR Threshold1	pos2-carthres1	Validation ratio for lambda search with CAR.
23	CAR Threshold2	pos2-carthres2	Maximum 3D-position std-dev to start narrow-lane ambiguity search with CAR. (m)
24	Signal Selection Option GPS IIR-M	pos2-siggpsIIR-M	Defines the signal selection for GPS IIR-M. (0:L1C/A-L2P,1:L1C/A-L2C ,3:L1C/A-L2)
25	Signal Selection Option GPS IIF	pos2-siggpsIIF	Defines the signal selection for GPS IIF. (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)
26	Signal Selection Option GPS IIIA	pos2-siggpsIIIA	Defines the signal selection for GPS IIIA. (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)
27	Signal Selection Option for GAL	pos2-siggal	Defines the signal selection for Galileo. (0:E1-E5a,1:E1-E5b,2:E1-E5a-E5b,3:E1-E5a-E5b-E6)
28	Signal Selection Option QZS1/2	pos2-sigqzs1_2	Defines the signal selection for QZSS Block I and II (QZS02,03,04). (0:L1C-L5,1:L1C/A-L2C ,3:L1C-L2C-L5,4:L1C/A-L2C-L5)
29	Signal Selection Option QZS-2A	pos2-sigqzs2A	Defines the signal selection for QZSS Block IIA (QZS1R). (0:L1C-L5,1:L1C/A-L2C,2:L1C/B-L5,3:L1C-L2C-L5,4:L1C/A-L2C-L5,5:L1C/B-L2C-L5)
30	Signal Selection Option QZS-3	pos2-sigqzs3	Defines the signal selection for QZSS Block III (QZS05,06,07). (0:L1C-L5 ,2:L1C/B-L5)
31	Signal Selection Option BDS-2	pos2-sigbds2	Defines the signal selection for BDS-2. (0:B1I-B3I,1:B1I-B2I,2:B1I-B3I-B2I)

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No.	Item	Configuration File	Descriptions
32	Signal Selection Option BDS-3	pos2-sigbds3	Defines the signal selection for BDS-3. (0:B1-B3,1:B1C-B2a,2:B1I-B3I-B2a)
33	Chi-Squared Test Anomaly Disable Option	pos2-ign_chierr	Ignore chi-square error mode (0:off,1:on) When it's activated, it ignores chi-square error and continues single point positioning.
34	BeiDou-2 ISB Estimation	pos2-bds2bias	Estimates the inter-system bias (ISB) for BeiDou-2. (0:off,1:on)
35	Satellite Code Bias for PPP	pos2-pppsatcb	Defines the method used for satellite code bias correction. (0:auto,1:l6e,2:bia,3:dcb,4:rtcm3) -Auto : Selects the first available valid bias correction method in the order of: l6e, bia, dcb, rtcm3. -L6E : Uses L6E signal-based bias correction. -BIA : Applies BIA (Bias Information Analysis) method. -DCB : Uses Differential Code Bias data. -RTCM3 : Applies RTCM Version 3 standard bias correction.
36	Satellite Phase Bias for PPP	pos2-pppsatpb	Defines the method for selecting satellite phase bias. (0:auto,1:l6e,2:bia,3:fcb,4:rtcm3)
37	Uncorrected Code Bias	pos2-uncorrbias	0:use uncorrected code bias, 1: do not use uncorrected code bias, 2: do not use uncorreced satellite bias
38	Bias Validity	pos2-maxbiasdt	Defines the Bias SINEX code bias timeout in
39	Missing number	-	-
40	Processing Interval	pos2-tint	Defines the parameter used to relax the threshold when detecting and removing outliers based on the standard deviation of PPP residual observation errors, taking into account missing data periods. (s)
41	Clock Estimation	pos2-clkopt	Estimation option of receiver clock bias with random walk.(0:off,1:on) If pos1-dynamics is set to ins, this setting is used.
42	Clock Drift Estimation	pos2-clkdopt	Estimation option of receiver clock drift with random walk. (0:off,1:on) If pos1-dynamics is set to ins, this setting is used.
43	IMU Measurement Interval	pos2-insint	Defines the measurement interval for IMU.(s) This value is used to determine the update interval and compute the process noise for INS inertial navigation.

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No.	Item	Configuration File	Descriptions
44	INS Data Outage Detection Threshold	pos2-insreset	Defines the timeout threshold for INS sensor data. When the outage duration exceeds this value, the position state is initialized with the stand-alone solution.
45	Orbit Dynamics Model Initialization Threshold	pos2-maxdist	Defines the initialization threshold for the orbit dynamics model. If the distance between the current position solution and the model exceeds this value, the model is reinitialized.
46	Observation-Excluded Object	pos2-maskobj	Defines the objects to be excluded from observation in the azimuth/elevation mask definition. For example, "STVX" excludes all objects.
47	URA Ratio	stats-uraratio	Defines the URA(User Range Accuracy) ratio.
48	Initial Standard Deviation : Receiver Attitude	stats-stdatt	Defines the initial standard deviation of the receiver attitude. (deg)
49	Initial Standard Deviation : Receiver Velocity	stats-stdvel	Defines the initial standard deviation of the receiver velocity. (m/s)
50	Initial Standard Deviation : Receiver Position	stats-stdpos	Defines the initial standard deviation of the receiver position. (m)
51	Initial Standard Deviation : Receiver Accelerometer Bias	stats-stdaccb	Defines the initial standard deviation of the receiver accelerometer bias. (micro-g)
52	Initial Standard Deviation : Receiver Gyro Bias	stats-stdgyrob	Defines the initial standard deviation of the receiver gyro bias. (deg/h)
53	Initial Standard Deviation : Receiver Clock Offset	stats-stdclko	Defines the initial standard deviation of the receiver clock offset. (m)
54	Initial Standard Deviation : Receiver Clock Drift	stats-stdclkd	Defines the initial standard deviation of the receiver clock drift. (m/s)
55	Process Noise : Receiver Clock Drift	stats-prnclkd	Defines the process noise deviation of the receiver clock drift. (m)
56	Process Noise : Receiver GPS Clock Offset	stats-prnclk_gps	Defines the process noise deviation of the receiver GPS clock offset. (m)
57	Process Noise : Receiver GLO Clock Offset	stats-prnclk_glo	Defines the process noise deviation of the receiver GLONASS clock offset. (m)

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No.	Item	Configuration File	Descriptions
58	Process Noise : Receiver GAL Clock Offset	stats-prnclk_gal	Defines the process noise deviation of the receiver Galileo clock offset. (m)
59	Process Noise : Receiver QZS Clock Offset	stats-prnclk_qzs	Defines the process noise deviation of the receiver QZSS clock offset. (m)
60	Process Noise : Receiver BDS Clock Offset	stats-prnclk_bds	Defines the process noise deviation of the receiver BeiDou clock offset. (m)
61	Process Noise : Receiver IRN Clock Offset	stats-prnclk_irn	Defines the process noise deviation of the receiver IRNSS clock offset. (m)
62	Process Noise : Receiver Accelerometer Noise	stats-prnaccn	Defines the process noise deviation of the receiver accelerometer noise. (micro-g/Hz)
63	Process Noise : Receiver Gyro Noise	stats-prngyron	Defines the process noise deviation of the receiver gyro noise. (deg/h)
64	Process Noise : Receiver Accelerometer Bias	stats-prnaccb	Defines the process noise deviation of the receiver accelerometer bias. (micro-g/Hz)
65	Process Noise : Receiver Gyro Bias	stats-prngyrob	Defines the process noise deviation of the receiver gyro bias. (deg/h)
66	Detecting Clock Jumps Threshold	stats-clkjump	Defines the threshold for detecting clock jumps(s)
67	Lever Arm Between GNSS and IMU	ant1-antimux	Defines the lever arm between the GNSS and IMU. (m)
68		ant1-antimuy	
69		ant1-antimuz	
70	IMU Rotation	ant1-rotimux	Defines the rotation of the IMU. (deg)
71		ant1-rotimuy	
72		ant1-rotimuz	
73	RTCM Option	misc-rtcmopt	Specify RTCM3 options for MADOCA-SSR as follows. -IGNORE_MCSSR_ALERT : Ignore the MADOCA-SSR Alert Flag -MCSSR_PRN={PRN} : Only decode MADOCA-SSR for the specified PRN -DIS_MCSSR_CB : Do not use madoca-ssr Code Bias
74	File Path of fcb Product	file-fcbfile	Input the file path of the fcb product file.
75	File Path of BIA Product	file-biafile	Input the file path of the BIA Product(Bias SINEX) file.

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No.	Item	Configuration File	Descriptions
76	Azimuth/Elevation Mask Definition File	file-maskfile	Specifies the path to the azimuth/elevation mask definition file.
77	Satellite Exclusion Configuration File	file-exclsatsfile	Specifies the path to the satellite exclusion configuration file.
78	PPP Kalman Filter Output File	file-pppkf	Specify the file path for the PPP Kalman filter state vector and covariance output.[*.kf] * Will be replaced with the actual output file name.

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Table 2 Configuration options added in MALIB for RTKRCV

No.	Item	Configuration File	Descriptions
1	Input Stream Type	inpstr4-type	Define the input stream type. (inpstr1 to inpstr4) (0:off,1:serial,2:file,3:tcpsvr,4:tcpcli,6:ntripcli,7:ftp,8:http,10:udpsvr,11:udpcli) -OFF : No input stream is used. -SERIAL : Input is received via a serial port. -FILE : Input is read from a file. -TCPSVR : Input is received from a TCP server. -TCPCLI : Input is received from a TCP client. -NTRIPCLI : Input is received from an NTRIP client (typically used for GNSS correction data). -FTP : Input is received via FTP. -HTTP : Input is received via HTTP. -UDPSVR : Input is received from a UDP server. -UDPCLI : Input is received from a UDP client.
2	Input Stream Path	inpstr4-path	Define the input stream path for stream. (inpstr4) The format depends on the selected stream type. Stream path formats: - SERIAL : port[:bit_rate[:byte[:parity(n|o|e)[:stopb[:fctr(off|on)]]]]][#port*1] - FILE : path[::T[::+offset][::xspeed]] - TCPSVR : :port - TCPCLI : addr:port - NTRIPCLI : [user[:passwd]]@addr:port/mntpnt - FTP : user:passwd@addr/path[::T=poff,tint,off,rint] - HTTP : addr/path[::T=poff,tint,off,rint] - UDPSVR : :port - UDPCLI : addr:port
3	Input Stream Format	inpstr4-format	Define the data format of the input stream (inpstr4). This setting determines how GNSS data is interpreted. (0:rtcm2,1:rtcm3,2:oem4,3:oem3,4:ubx,5:ss2,6:hemis,7:skytraq,8:javad,9:nvs,10:binex,11:rt17,12:sbf,14:sp3,16:l6) - RTCM2 : RTCM Version 2 format (GNSS correction data) - RTCM3 : RTCM Version 3 format (GNSS correction data) - OEM4 : NovAtel OEM4 binary format - OEM3 : NovAtel OEM3 binary format - UBX : u-blox proprietary binary format - SS2 : NovAtel SuperStar II format - HEMIS : Hemisphere GNSS binary format - SKYTRAQ : SkyTraq binary format - JAVAD : JAVAD GNSS binary format - NVS : NVS binary format - BINEX : BINEX (Binary Exchange Format for GNSS data) - RT17 : Trimble RT17 format - SBF : Septentrio Binary Format - SP3 : Standard Product 3 format (precise satellite ephemeris) - L6 : L6E format

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No.	Item	Configuration File	Descriptions
4	Input Stream SSR	inpstr1-ssr inpstr2-ssr inpstr3-ssr inpstr4-ssr	Defines the type of SSR (State Space Representation) data input for each stream (inpstr1 to inpstr4). This setting specifies how SSR data is interpreted depending on the source. -MAIN : Receive SSR data from the main antenna -SUB : Receive SSR data from the sub antenna -OFF: SSR input is disabled Note: Each of MAIN and SUB can be assigned only once across all streams (inpstr1 to inpstr4).
5	Logging Stream Type	logstr4-type	Define the logging stream type (logstr4). This setting determines how solution and observation data are logged. (0:off, 1:serial, 2:file, 3:tcpsvr, 4:tcpcli, 5:ntripsvr, 9:ntripcas, 10:udpsvr, 11:udpcli) -OFF : No logging is performed. -SERIAL : Data is logged from a serial port. -FILE : Data is logged to a file. -TCPSVR : Data is logged from a TCP server. -TCPCLI : Data is logged from a TCP client. -NTRIPSVR : Data is logged to an NTRIP server (used to broadcast GNSS correction data). -NTRIPCAS : Data is logged to an NTRIP caster (used to relay GNSS correction streams). -UDPSVR : Data is logged from a UDP server. -UDPCLI : Data is logged from a UDP client.
6	Logging Stream Path	logstr4-path	Define the logging stream path. (logstr4) The format depends on the selected stream type. Stream path formats: - SERIAL : port[:bit_rate[:byte[:parity(n|o|e)[:stopb[:fctr(off|on)]]]]][#port*1] - FILE : path[::T[::+offset][::xspeed]] - TCPSVR : :port - TCPCLI : addr:port - NTRIPSVR : :passwd@addr:port/mntpnt[:str*2] - NTRIPCAS : [user[:passwd]]@:port/mntpnt[:str*2] - UDPSVR : :port - UDPCLI : addr:port
7	Transition Threshold for Dead Reckoning	misc-maxdrtr	Defines the transition threshold for switching to INS dead reckoning. (s)

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Table 3 Startup arguments added in MALIB

No.	Item	Startup arguments (rtkrcv)	Description
1	RTCM data playback start time	-rst ds ts (ds=y/m/d ts=h:m:s)	raw/rtcm data start timeSet this if the PC time and data time do not match when checking playback operation.
2	Output PPP kf State/Covariance	-opkf level	Output PPP Kalman filter state/covariance. (0:off,1:states,2:states + diagonal covariance, 3:states + diagonal covariance + full covariance*) [0] *only at final epoch

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3.3. Sample Settings

A sample configuration file for RNX2RTKP and RTKRCV.

```
#Sample Configuration Settings for RNX2RTKP and RTKRCV
# Setting1
pos1-posmode       =ppp-kine            # Positioning Mode
(0:single,1:dgps,2:kinematic,3:static,4:movingbase,5:fixed,6:ppp-kine,7:ppp-static,8:ppp-fixed)
pos1-frequency     =l1+2+3+4            # Frequencies
(1:l1,2:l1+2,3:l1+2+3,4:l1+2+3+4,5:l1+2+3+4+5)
pos1-soltype       =forward             # Filter Type (0:forward,1:backward,2:combined)
pos1-elmask        =10                  # Elevation Mask (deg)
pos1-snrmask_r     =off                 # SNR Mask : Target (0:off,1:on)
pos1-snrmask_b     =off                 # SNR Mask : Base-Station (0:off,1:on)
## SNR Mask : Threshold
##                    5,  15,  25,  35,  45,  55,  65,  75,  85 #EL(deg)
pos1-snrmask_L1    = 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
pos1-snrmask_L2    = 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
pos1-snrmask_L5    = 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
##
pos1-dynamics      =off                 # Receiver Dynamics (0:off,1:dynamics,2:ins)
pos1-tidecorr      =on                  # Earth Tides Correction (0:off,1:on,2:otl)
pos1-ionoopt       =est-stec            # Ionosphere Correction (0:off,1:brdc,2:sbas,3:dual-
freq,4:est-stec,5:ionex-tec,6:qzs-brdc)
pos1-tropopt       =est-ztd             # Troposphere Correction (0:off,1:saas,2:sbas,3:est-
ztd,4:est-ztdgrad)
pos1-sateph        =brdc+ssrapc         # Satellite Ephemeris/Clock
(0:brdc,1:precise,2:brdc+sbas,3:brdc+ssrapc,4:brdc+ssrcom)
pos1-posopt1       =on                  # Satellite PCV (0:off,1:on)
pos1-posopt2       =on                  # Receiver PCV (0:off,1:on)
pos1-posopt3       =on                  # Phase Windup (0:off,1:on,2:precise)
pos1-posopt4       =on                  # Reject Eclipse (0:off,1:on)
pos1-posopt5       =off                 # RAIM FDE (0:off,1:on)
pos1-posopt6       =off                 # Day-boundary jump (0:off,1:on)
pos1-exclsats      =                    # Excluded Satellites (prn ...)
pos1-navsys        =61                  # Navigation System
(1:gps+2:sbas+4:glo+8:gal+16:qzs+32:bds+64:navic)
pos1-subopt        =off                 # Substitute PPP Option (0:off,1:brdc,4:rt)
pos1-drint         =0.0                 # Dead Reckoning Interval (s)
## SNR Waiting : Threshold
##                    sig=   5,  10,  15,  20,  25,  30,  35,  40,  45,  50,  55,  60,  65,  70,
75,  80,  85 #EL(deg)
pos1-snrwthres_gps= 1C= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gps= 2P= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gps= 2L= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gps= 5Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_glo= 1C= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_glo= 2P= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gal= 1C= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gal= 5Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gal= 7Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_gal= 6C= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_qzs= 1L= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_qzs= 5Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_qzs= 1C= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_qzs= 1E= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_qzs= 2L= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_bds= 2Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_bds= 6Q= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
pos1-snrwthres_bds= 5P= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
```


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```
0.0, 0.0, 0.0
pos1-snrwthres_bds= 7I= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0
## SNR Waiting : Margin
pos1-snrwmargin_gps=0.0                 # GPS
pos1-snrwmargin_glo=0.0                 # GLONASS
pos1-snrwmargin_gal=0.0                 # Galileo
pos1-snrwmargin_qzs=0.0                 # QZSS
pos1-snrwmargin_bds=0.0                 # BeiDou
## SNR Waiting : Error Ratio
pos1-snrweratio_gps=0.0                 # GPS
pos1-snrweratio_glo=0.0                 # GLONASS
pos1-snrweratio_gal=0.0                 # Galileo
pos1-snrweratio_qzs=0.0                 # QZSS
pos1-snrweratio_bds=0.0                 # BeiDou

# Setting2
pos2-ionocorr      =off                 # MADOCA-PPP Ionospheric Correction (0:off,1:on)
pos2-armode        =off                 # Integer Ambiguity Resolution
(0:off,1:continuous,2:instantaneous,3:fix-and-hold)
pos2-arsys         =57                  # GNSS Used for PPP-AR (1:gps+8:gal+16:qzs+32:bds)
pos2-car           =on                  # Cascading Ambiguity Resolution (0:off,1:on)
pos2-arthres       =3                   # Min Ratio to Fix Ambiguity
pos2-arthres1      =0.9999              # Min Confidence to Fix Ambiguity
pos2-arthres2      =0.25                # Max FCB to Fix Ambiguity
pos2-arthres3      =0.1                 # Max WL Deviation to Fix Ambiguity
pos2-arthres4      =0.05                # Max NL Deviation to Fix Ambiguity
pos2-carthres1     =1.5                 # Validation ratio for lambda search with CAR
pos2-carthres2     =1.5                 # Maximum 3D-position std-dev to start narrow-lane
ambiguity search with CAR (m)
pos2-arlockcnt     =0                   # Min Lock to Fix Ambiguity
pos2-arelmask      =15                  # Min Elevation to Fix Ambiguity (deg)
pos2-arminfix      =10                  # Min Fix to Hold Ambiguity
pos2-armaxiter     =99                  # Max AR Iterations
pos2-elmaskhold    =0                   # Min Elevation to Hold Ambiguity (deg)
pos2-aroutcnt      =5                   # Outage to Reset Ambiguity
pos2-maxage        =30                  # Max Age of Differential (s)
pos2-slipthres     =0.15                # Outage to Slip Thres for Geometry-Free LC (m)
pos2-rejionno      =100                 # Reject Threshold of innovation
pos2-rejgdop       =30                  # Reject Threshold of GDOP
pos2-niter         =1                   # Number of Iteration
pos2-baselen       =0                   # Baseline Length (m)
pos2-basesig       =0                   # Baseline length standard deviation (m)
## Signal Selection Option
pos2-siggpsIIR-M   =L1C/A-L2P           # GPS BLOCK IIR-M (0:L1C/A-L2P,1:L1C/A-
L2C           ,3:L1C/A-L2)
pos2-siggpsIIF     =L1C/A-L2P-L5        # GPS BLOCK IIF   (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-
L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)
pos2-siggpsIIIA    =L1C/A-L2P-L5        # GPS BLOCK IIIA  (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-
L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)
pos2-siggal        =E1-E5a-E5b-E6       # Galileo (0:E1-E5a,1:E1-E5b,2:E1-E5a-E5b,3:E1-E5a-E5b-
E6)
pos2-sigqzs1_2     =L1C-L5-L2C          # QZSS Block I and II (QZS02,03,04) (0:L1C-L5,1:L1C/A-
L2C           ,3:L1C-L5-L2C,4:L1C/A-L2C-L5)
pos2-sigqzs2A      =L1C-L5-L2C          # QZSS Block IIA (QZS1R)            (0:L1C-L5,1:L1C/A-
L2C,2:L1C/B-L5,3:L1C-L5-L2C,4:L1C/A-L2C-L5,5:L1C/B-L2C-L5)
pos2-sigqzs3       =L1C-L5              # QZSS Block III (QZS05,06,07)      (0:L1C-
L5            ,2:L1C/B-L5)
pos2-sigbds2       =B1I-B3I-B2I            #    BeiDou-2    (0:B1I-B3I,1:B1I-B2I,2:B1I-B3I-B2I)
pos2-sigbds3       =B1I-B3I-B2a         # BeiDou-3 (0:B1I-B3I,1:B1I-B2a,2:B1I-B3I-B2a)
##
pos2-ign_chierr    =on                  # Chi-Squared Test Error Ignore Option (0:off,1:on)
pos2-bds2bias      =off                 # BeiDou-2 ISB Estimation (0:off,1:on)
pos2-pppsatcb      =auto                # Satellite Code Bias Option
(0:auto,1:l6e,2:bia,3:dcb,4:rtcm3)
pos2-pppsatpb      =auto                # Satellite Phase Bias Option
(0:auto,1:l6e,2:bia,3:fcb,4:rtcm3)
pos2-uncorrbias    =use                 # Uncorrected Code Bias (0:use,1:notuse,2:notuse-
uncorrsatbias)
pos2-maxbiasdt     =0                   # Bias Validity Time Threshold (s)

pos2-tint          =0.1                 # Processing Interval(s)
pos2-clkopt        =off                 # Clock Estimation (0:off,1:on)
pos2-clkdopt       =off                 # Clock Drift Estimation (0:off,1:on)
pos2-insint        =0.005               # INS Measurement Interval(s)
pos2-insreset      =0.0                 # INS Data Outage Detection Threshold(s)
pos2-maxdist       =600                 # Orbit Dynamics Model Initialization Threshold(m)
pos2-maskobj       =                    # Observation Excluded Object
(S:Structure,T:Terrain,V:Vegetation,X:eXtra)

# Output
out-solformat      =llh                 # Solution Format (0:llh,1:xyz,2:enu,3:nmea)
```


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```
out-outhead         =on                 # Output Header (0:off,1:on)
out-outopt          =on                 # Output Processing Options (0:off,1:on)
out-outvel          =off                # Output Velocity (0:off,1:on)
out-timesys         =gpst               # Time System (0:gpst,1:utc,2:jst)
out-timeform        =hms                # Time Format (0:tow,1:hms)
out-timendec        =3                  # Time Format of Decimals
out-degform         =deg                # Latitude/Longitude Format (0:deg,1:dms)
out-fieldsep        =                   # Field Separator
out-outsingle       =off                # Single Solution Output (0:off,1:on)
out-maxsolstd       =0                  # Maximum Solution Standard Deviation (m)
out-height          =ellipsoidal        # Type of Height (0:ellipsoidal,1:geodetic)
out-geoid           =internal           # Geoid Model
(0:internal,1:egm96,2:egm08_2.5,3:egm08_1,4:gsi2000)
out-solstatic       =all                # Solution for Static Mode (0:all,1:single)
out-nmeaintv1       =0                  # NMEA Interval (s) RMC/GGA
out-nmeaintv2       =0                  # NMEA Interval (s) GSA/GSV
out-outstat         =state              # Output Solution Status (0:off,1:state,2:residual)

# Statistics
stats-eratio1       =100                # Code/Carrier-Phase Error Rate L1
stats-eratio2       =100                # Code/Carrier-Phase Error Rate L2
stats-errphase      =0.003              # Carrier-Phase Error (m)
stats-errphaseel    =0.003              # Carrier-Phase Error/sin (m)
stats-errphasebl    =0                  # Carrier-Phase Error/Baseline (m/10km)
stats-errdoppler    =1                  # Doppler Frequency (Hz)
stats-uraratio      =1.0                # URA Ratio
## Initial Standard Deviation
stats-stdbias       =30                 # Carrier-Phase Bias (m)
stats-stdiono       =0.03               # Ionospheric Delay (m)
stats-stdtrop       =0.3                # Tropospheric Delay (m)
stats-stdatt        =1                  # Receiver Attitude (deg)
stats-stdvel        =0.1                # Receiver Velocity (m/s)
stats-stdpos        =60                 # Receiver Position (m)
stats-stdaccb       =1000               # Receiver Accelerometer Bias (micro-g)
stats-stdgyrob      =0.01               # Receiver Gyroscope Bias (deg/h)
stats-stdclko       =60                 # Receiver Clock Offset (m)
stats-stdclkd       =0.05               # Receiver Clock Drift (m/s)
## Process Noise
stats-prnaccelh     =10                 # Receiver Accel Horizontal (m/s^2)
stats-prnaccelv     =10                 # Receiver Accel Vertical (m/s^2)
stats-prnbias       =0.0001             # Carrier-Phase Bias (m)
stats-prniono       =0.01               # Ionospheric Delay (m)
stats-prntrop       =0.0001             # Zenith Tropospheric Delay (m)
stats-prnpos        =0                  # Static Position (m)
stats-prnifb        =1e-5               # Inter Frequency Bias (m)
stats-prnclkd       =3                  # Receiver Clock Drift (m/sqrt(s))
stats-prnclk_gps    =3                  # Receiver GPS Clock Offset(m)
stats-prnclk_glo    =3                  # Receiver GLONASS Clock Offset(m)
stats-prnclk_gal    =3                  # Receiver Galileo Clock Offset(m)
stats-prnclk_qzs    =3                  # Receiver QZSS Clock Offset(m)
stats-prnclk_bds    =3                  # Receiver BeiDou-3 Clock Offset(m)
stats-prnclk_irn    =3                  # Receiver IRNSS Clock Offset(m)
stats-prnclk_bds2   =3                  # Receiver BeiDou-2 Clock Offset(m)
stats-prnaccn       =24.8               # Receiver Accelerometer Noise(micro-g/Hz)
stats-prngyron      =0.2                # Receiver Gyroscope Noise (deg/h)
stats-prnaccb       =0.0044             # Receiver Accelerometer Bias (micro-g/Hz)
stats-prngyrob      =2.7                # Receiver Gyroscope Bias (deg/h)
##
stats-clkjump       =0.0005             # Detecting Clock Jumps threshold (s)
stats-clkstab       =5e-12              # Satellite Clock Stability (s/s)

# Positions
## Antenna1
ant1-postype        =llh                # Position Type
(0:llh,1:xyz,2:single,3:posfile,4:rinexhead,5:rtcm,6:raw)
ant1-pos1           =90                 # Position 1 (deg|m)
ant1-pos2           =0                  # Position 2 (deg|m)
ant1-pos3           =-6335367.6285      # Position 3 (m|m)
ant1-anttype        =*                  # Antenna Type
ant1-antdele        =0                  # Delta-E (m)
ant1-antdeln        =0                  # Delta-N (m)
ant1-antdelu        =0                  # Delta-U (m)
ant1-antimux        =0                  # lever arm between GNSS and IMU Body-X (m)
ant1-antimuy        =0                  # lever arm between GNSS and IMU Body-Y (m)
ant1-antimuz        =0                  # lever arm between GNSS and IMU Body-Z (m)
ant1-rotimux        =0                  # IMU Rotation Body-X(deg)
ant1-rotimuy        =0                  # IMU Rotation Body-Y(deg)
ant1-rotimuz        =0                  # IMU Rotation Body-Z(deg)
## Antenna2
ant2-postype        =llh                # Position Type
(0:llh,1:xyz,2:single,3:posfile,4:rinexhead,5:rtcm,6:raw)
ant2-pos1           =90                 # Position 1 (deg|m)
```


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```
ant2-pos2          =0                  # Position 2 (deg|m)
ant2-pos3          =-6335367.6285      # Position 3 (m|m)
ant2-anttype       =                   # Antenna Type
ant2-antdele       =0                  # Delta-E (m)
ant2-antdeln       =0                  # Delta-N (m)
ant2-antdelu       =0                  # Delta-U (m)
ant2-maxaveep      =0                  # Max Average Epochs
ant2-initrst       =off                # Init Reset(0:off,1:on)

# Miscellaneous
misc-timeinterp    =off                # Time Interpolation of Base Station Observation Data
(0:off,1:on)
misc-sbasatsel     =0                  # SBAS Satellite Selection (0:all)
misc-rnxopt1       =                   # RINEX Option (Target)
misc-rnxopt2       =                   # RINEX Option (Base)
misc-pppopt        =                   # PPP Option
misc-rtcmopt       =                   # RTCM Option

# Files
file-satantfile    =./igs14_2239.atx   # Satellite Antenna PCV File ANTEX
file-rcvantfile    =./igs14_2239.atx   # Receiver Antenna PCV File ANTEX or NGS PCV
file-staposfile    =                   # Station Position File
file-geoidfile     =                   # Geoid Data File
file-ionofile      =                   # IONEX-TEC Data File
file-dcbfile       =                   # DCB Data File
file-eopfile       =                   # EOP Data File
file-blqfile       =                   # Ocean Loading BLQ Format File
file-fcbfile       =                   # FCB ProductFile
file-biafile       =                   # BIA ProductFile
file-tempdir       =                   # FTP/HTTP Local Directory
file-solstatfile   =                   # Stat Output File
file-tracefile     =                   # Trace Output File
file-maskfile      =                   # Azimuth/Elevation Mask Definition File
file-exclsatsfile  =                   # Satellite Exclusion Configuration File
file-pppkf         =                   # PPP Kalman Filter Output File
```


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```
#Sample Configuration Settings for RTKRCV-Specific
# Console
console-passwd     =                    # Password
console-timetype   =gpst                # Time Type (0:gpst,1:utc,2:jst,3:tow)
console-soltype    =deg                 # Solution Type (0:dms,1:deg,2:xyz,3:enu,4:pyl)
console-solflag    =off                 # Solution Flag (0:off,1:std+2:age/ratio/ns)

# Input Stream
## Type (0:off,1:serial,2:file,3:tcpsvr,4:tcpcli,6:ntripcli,7:ftp,8:http,10:udpsvr,11:udpcli)
inpstr1-type       =off
inpstr2-type       =off
inpstr3-type       =off
inpstr4-type       =off
## Path
inpstr1-path       =
inpstr2-path       =
inpstr3-path       =
inpstr4-path       =
## Format
(0:rtcm2,1:rtcm3,2:oem4,3:oem3,4:ubx,5:ss2,6:hemis,7:skytraq,8:javad,9:nvs,10:binex,11:rt17,12:s
bf,14:sp3,15:l6,20:stat)
inpstr1-format     =
inpstr2-format     =
inpstr3-format     =
inpstr4-format     =
## SSR (0:main,1:sub,-1:off)
inpstr1-ssr        =off
inpstr2-ssr        =mai
inpstr3-ssr        =off
inpstr4-ssr        =off
## NMEA
inpstr2-nmeareq    =off                 # Request (0:off,1:latlon,2:single)
inpstr2-nmealat    =0                   # Latitude (deg)
inpstr2-nmealon    =0                   # Longitude (deg)

# Output Stream
## Type (0:off,1:serial,2:file,3:tcpsvr,4:tcpcli,5:ntripsvr,9:ntripcas,10:udpsvr,11:udpcli)
outstr1-type       =file
outstr2-type       =file
## Path
outstr1-path       =sol_%Y%m%d%h%M.pos
outstr2-path       =sol_%Y%m%d%h%M.stat
## Format (0:llh,1:xyz,2:enu,3:nmea,4:stat)
outstr1-format     =llh
outstr2-format     =stat

# Log Stream
## Type (0:off,1:serial,2:file,3:tcpsvr,4:tcpcli,5:ntripsvr,9:ntripcas,10:udpsvr,11:udpcli)
logstr1-type       =off
logstr2-type       =off
logstr3-type       =off
logstr4-type       =off
## Path
logstr1-path       =
logstr2-path       =
logstr3-path       =
logstr4-path       =

# Miscellaneous
misc-maxdrtr       =0.0                 # Transition Threshold for Dead Reckoning (s)
misc-svrcycle      =10                  # Processing Cycle/ Buffer Size (ms)
misc-timeout       =180000              # Timeout (ms)
misc-reconnect     =10000               # Re-Connect Interval (ms)
misc-nmeacycle     =5000                # NMEA Cycle/ File Swap Margin (ms)
misc-buffsize      =32768               # Solution Buffer/Log Size (bytes)
misc-navmsgsel     =all                 # Navigation Message Selection
(0:all,1:rover,2:base,3:corr)
misc-proxyaddr     =                    # HTTP/NTRIP Proxy
misc-fswapmargin   =30                  # File Swap Overlap Time (s)
```


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Appendix A : Command References

A.1 LCLCMBPP

SYNOPSIS

lclcmbpp [options] file

DESCRIPTION

Generate local combination data as a post-processing step.

OPTIONS

-ts y/m/d hh:mm:ss start time
(y=year,m=month,d=date,hh=hour,mm=minite,ss=seconds in GPST)*
-te y/m/d hh:mm:ss end time [ts+86400s]
-grid file generate grid info file. *
-sta file generate station info file. *
-interp mode interpolation mode.[0] (0:planer fitting,1:weighted average of distance)
-extp enable extrapolation check on planer fitting mode. (only planer fitting)[off]
-idist dist max ionospheric delay distance (km) from target to station.[50]
-tdist dist max tropospheric delay distance (km) from target to station.[100]
-ni ni maxth fail station detection parameter.[ni=1,maxth=1.0]
(ni=itaraion number(1-5), maxth=max threshold)
up to 5 iterations, each with its own threshold.
-eratio ratio residual error ratio.[0.1]
-out file output file(.rtcm3)
-t level debug trace level (0:off) [0]
File input STAT file *
(* indicates required options)

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A.2 LCLCMBRT

SYNOPSIS

lclcmbpp [options] file

DESCRIPTION

Generate local combination data as a realtime step.

OPTIONS

-instr file	input stream option file *
-grid file	generate grid option file. *
-sta file	generate station option file. *
-idist dist	max ionospheric delay distance (km) from target to station.[50]
-tdist dist	max tropospheric delay distance (km) from target to station.[100]
-out path	output stream of combined data *
-log path	log stream of combined data
-p port	port number for monitor console
-t level	debug trace level (0:off) [0]
-rep	enables replay mode based on input data timestamps.

(* indicates required options)

COMMANDS

stat [-r]

Show stream status. repeated with r. stopped by ctrl C.

path

Show stream paths

enable sta

Enable stream of station sta

disable sta

Disable stream of station sta

shutdown

Shutdown lclcmbrt

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logout

Logout monitor console

help|?

Show help

In the stream status, the fields indicate as follows

NO	stream number
STA	station name (OUT:combined data)
F	stream formmat (S:STAT,R:RTCM3)
STAT	number of stat data messsages
KBYTE	input/output data amount (KB)
KBPS	input/output data rate (KB/s)
CS	connect status(-:close,W:wait,C:connect,E:error)
ES	enable/disable status(E:Enable,D:Disable)
STR_STATUS	status/error messages

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A.3 RECVBIAS

SYNOPSIS

recvbias [options] file

DESCRIPTION

It reads RINEX OBS/NAV, IONEX TEC data, and satellite code biases, and generates receiver code biases. The generated receiver code biases, as well as input satellite code biases (only when Bias SINEX format files are input), are saved to a file specified with option -out. All file paths can include keywords replaced by date and time. RINEX OBS/NAV and IONEX TEC files can include wildcards (*) to be expanded to multiple files.

OPTIONS

-td y/m/d	date (y=year,m=month,d=date in GPST) *
-ts tspan	time span (days) [1]
-nav file	RINEX NAV file *
-tec file	IONEX TEC file *
-scb file	satellite code bias file(.bia or .l6 or .rtcm3) *
-pos x,y,z	receiver position(ecef-x,y,z) *
-el elmask	elevation mask(deg)[30]
-sta staname	station name[RINEX MARKER NAME]
-sig sx[,sx...]	signal selection(s=G:GPS,J:QZS,C:BDS x=0-2)
	G0(L1C/A-L2P),G1(L1C/A-L2C),G2(L1C/A-L5)
	E0(E1-E5a),E1(E1-E5b)
	J0(L1C-L5),J1(L1C/A-L2C),J2(L1C/B-L5)
	C0(B1-B3),C1(B1C-B2a),C2(B1I-B2a)
-allbias	outputs receiver code biases for each satellite
-out file	output satellite/receiver code bias file [stdout]
-t level	debug trace level (0:off) [0]
File	RINEX OBS file *

(* indicates required options)

The program supports the following input file formats

- Bias SINEX format (.bia)
- L6 archive (.l6)
- RTCM3 (.rtcm3)

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keywords replaced in file path

%Y -> yyyy	year (4 digits)	(2000-2099)
%y -> yy	year (2 digits)	(00-99)
%m -> mm	month	(01-12)
%d -> dd	day of month	(01-31)
%h -> hh	hours	(00-23)
%H -> a	hour code	(a-x)
%M -> mm	minutes	(00-59)
%n -> ddd	day of year	(001-366)
%W -> wwww	gps week	(0001-9999)
%D -> d	day of gps week	(0-6)

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A.4 RTKRCV

SYNOPSIS

rtkrcv [-s][-p port|-d dev][-o file][-t level]

DESCRIPTION

A console-based real-time positioning application by MALIB. To start or stop RTK server, to configure options or to print solution/status, login a console and input commands. As default, stdin/stdout are used for the console. Use -p option for network login with telnet protocol. To show the available commands, type ? or help on the console. The initial processing options are loaded from default configuration file rtkrcv.conf. To change the file, use -o option. To configure the processing options, edit the configuration file or use set, load or save command on the console. To shut down the program, use shutdown command on the console or send the USR2 signal to the process. For configuration file, refer B.3.

OPTIONS

-s	start RTK server on program startup
-p port	port number for telnet console
-m port	port number for monitor stream
-d dev	terminal device for console
-o file	configuration file
-w pwd	login password for remote console (\"\": no password)
-r level	output solution status file (0:off,1:states,2:residuals)
-t level	debug trace level (0:off,1-5:on)
-opkf level	output PPP kalman filter state/covariance (0:off,1:states,2:states+dcov,3:states+dcov+ccov)
-sta sta	station name for receiver dcb
-v|-ver	print version
-rst ds ts	start day/time (ds=y/m/d ts=h:m:s) [raw/rtcm data start time]
-ignsync	ignore time sync for input files [sync]

COMMANDS

start

Start RTK server. No need the command if the program runs with -s option.

stop

Stop RTK server.

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restart

Restart RTK server. If the processing options are set, execute the command to enable the changes.

solution [cycle]

Show solutions. Without option, only one solution is shown. With option, the solution is displayed at intervals of cycle (s). To stop cyclic display, send break (ctr-C).

status [cycle]

Show RTK status. Use option cycle for cyclic display.

satellite [cycle]

Show satellite status. Use option cycle for cyclic display.

observ [cycle]

Show observation data. Use option cycle for cyclic display.

navidata [cycle]

Show navigation data. Use option cycle for cyclic display.

stream [cycle]

Show stream status. Use option cycle for cyclic display.

ssr [cycle]

Show ssr corrections. Use option cycle for cyclic display.

error

Show error/warning messages. To stop messages, send break (ctr-C).

option [opt]

Show the values of processing options. Without option, all options are displayed. With option, only pattern-matched options are displayed.

set opt [val]

Set the value of a processing option to val. Without option val, prompt message is shown to input the value. The change of the processing option is not enabled before RTK server is restarted.

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load [file]

Load processing options from file. Without option, default file rtkrcv.conf is used.To enable the changes, restart RTK server.

save [file]

Save current processing options to file. Without option, default file rtkrcv.conf is used.

log [file|off]

Record console log to file. To stop recording the log, use option off.

help|? [path]

Show the command list. With option path, the stream path options are shown.

exit|ctr-D

Exit and logout console. The status of RTK server is not affected by the command.

shutdown

Shutdown RTK server and exit the program.

!command [arg…]

Execute command by the operating system shell. Do not use the interactive command.

Note

Short form of a command is allowed. In case of the short form, the command is distinguished according to header characters.

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A.5 RNX2RTKP

SYNOPSIS

rnx2rtkp [option …] file file […]

DESCRIPTION

Read RINEX OBS/NAV/GNAV/HNAV/CLK, SP3, SBAS message log, RTCM3, L6(MADOCA-PPP) files and compute receiver (rover) positions and output position solutions.

The first RINEX OBS file shall contain receiver (rover) observations. For the relative mode, the second RINEX OBS file shall contain reference (base station) receiver observations. At least one RINEX NAV/GNAV/HNAV file shall be included in input files. To use SP3 precise ephemeris, specify the path in the files. The extension of the SP3 file shall be .sp3 or .eph. L6 (MADOCA-PPP) files support clock/ephemeris corrections and ionospheric Multiple ionospheric correction files can be specified as input corrections. All of the input file paths can include wild-cards (*). To avoid command line deployment of wild-cards, use "…" for paths with wild-cards. Command line options are as follows ([]:default). With -k option, the processing options are input from the configuration file. In this case, command line options precede options in the configuration file. For the configuration file, refer B.3.

OPTIONS

-? Print help
-k file input options from configuration file [off]
-o file set output file [stdout]
-ts ds ts start day/time (ds=y/m/d ts=h:m:s) [obs start time]
-te de te end day/time (de=y/m/d te=h:m:s) [obs end time]
-ti tint time interval (sec) [all]
-p mode mode (0:single,1:dgps,2:kinematic,3:static,4:moving-base,5:fixed,6:ppp-kinematic,
7:ppp-static) [2]
-m mask elevation mask angle (deg) [15]
-sys s[,s...] nav system(s) (s=G:GPS,R:GLO,E:GAL,J:QZS,C:BDS,I:IRN) [G|R]
-f freq number of frequencies for relative mode (1:L1,2:L1+L2,3:L1+L2+L5) [2]
-v thres validation threshold for integer ambiguity (0.0:no AR) [3.0]
-b backward solutions [off]
-c forward/backward combined solutions [off]
-i instantaneous integer ambiguity resolution [off]
-h fix and hold for integer ambiguity resolution [off]

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-e output x/y/z-ecef position [latitude/longitude/height]
-a output e/n/u-baseline [latitude/longitude/height]
-n output NMEA-0183 GGA sentence [off]
-g output latitude/longitude in the form of ddd mm ss.ss [ddd.ddd]
-t output time in the form of yyyy/mm/dd hh:mm:ss.ss [ssss.ss]
-u output time in utc [gpst]
-d col number of decimals in time [3]
-s sep field separator [‘ ‘]
-r x y z reference (base) receiver ecef pos (m) [average of single pos]
-l lat lon hgt reference (base) receiver latitude/longitude/height (deg/m)
-ign_chierr ignore chi-square error mode [off]
-sta staname station name[RINEX MARKER NAME]
-sbfatt attfile septentrio ASCII GNSS attitude(AttEuler) file name
-sbfimu imufile septentrio ASCII IMU sensor(ExtSensorMeas) file name
-ubxins ubxfile u-blox F9 high precision sensor fusion GNSS file name
-subssr ssrfile substitute PPP correction SSR file name
-y level output solution status (0:off,1:states,2:residuals) [0]
-x level debug trace level (0:off) [0]
-opkf level output PPP kalman filter state/covariance
(0:off,1:states,2:states+dcov,3:states+dcov+ccov) [0]
-ver print version

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Appendix B : File Formats

B.1 Positioning solution file

DESCRIPTION

A positioning solution file is just a text file as output of RTKNAVI or RTKPOST. The file is separated to records or lines by CR/LF. Each records consists of fields. The following table shows the format of the positioning solution file.

Table 4 Format of the positioning solution file

No	Record/Field	Description
1	File header	The lines starting with "%" are header lines. The header lines contains some additional information or processing options as follows. % program : program version % inp file : Input file path % obs start: Observation data start time in GPS time % obs end: Observation data end time in GPS time % pos mode: Positioning mode option % freqs: Frequencies option % solution: Solution type option % elev mask: Elevation mask angle option % dynamics: Rec Dynamics option % tidecorr: Earth Tides Correction option % ionos opt: Ionospheric parameter estimation option % tropo opt: Tropospheric parameters estimation option % ephemeris: Satellite Ephemeris/Clock option % navi sys: Navigation System option % ar sys: Satellite systems to be used for integer ambiguity resolution % amb res: Integer ambiguity resolution options % val thres: Integer ambiguity validation option % antenna1: target antenna option
2	Field indicator	Field indicator starting with "%" line follows after File header. To recognize the field formats, MALIB uses these lines. Do not delete them.
3	Solution body	Solution body consists of the following fields. The field contents are varied according to the positioning options.
(1)	Time	The epoch time of the solution indicating the true receiver signal reception time (not indicates the time by receiver clock). The format is varied to the options. yyyy/mm/dd HH:MM:SS.SSS : Calendar time in GPST, UTC or JST, the time system is indicated in Field indicator WWWW SSSSSSS.SSS : GPS week and TOW (time of week) in seconds.

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No	Record/Field	Description
(2)	Receiver Position	The antenna or marker position estimated varied according to the positioning options. +ddd.ddddddddd +ddd.ddddddddd hhhh.hhhh : Latitude, longitude in degrees and height in m. Minus value means south latitude or west longitude. The height indicates ellipsoidal or geodetic according to the positioning options. +ddd mm ss.sss +ddd mm ss.sss hhhh.hhhh : Latitude, longitude in degree, minute and second and height in m. +xxxxxxxxx.xxxx +yyyyyyyyy.yyyy +zzzzzzzz.zzzz : X/Y/Z components of ECEF frame in m. +eeeeeeeee.eeee +nnnnnnnnn.nnnn +uuuuuuuuu.uuuu : E/N/U components of baseline vector in m. The local coordinate is referenced to the target position.
(3)	Quality flag (Q)	The flag which indicates the solution quality. 1 : Fixed, solution by carrier-based relative positioning and the integer ambiguity is properly resolved. 2 : Float, solution by carrier-based relative positioning but the integer ambiguity is not resolved. 3 : Reserved 4 : DGPS, solution by code-based DGPS solutions or single point positioning with SBAS corrections. 5 : Single, solution by single point positioning 6 : PPP, solution by Precise Point Positioning
(4)	Number of valid satellites (ns)	The number of valid satellites for solution estimation.
(5)	Standard deviations (sdn, sde, sdu, sdne, sdeu, sdun or sdx, sdy, sdz, sdxy, sdyz, sdzx)	The estimated standard deviations of the solution are computed assuming an a priori error model and error parameters defined by the positioning options. If out-solformat is set to 0:llh or 2:enu, the standard deviations are provided in the ENU (East, North, Up) coordinate system: sde, sdn, and sdu represent the east, north, and up components in meters. If out-solformat is set to 1:xyz, the standard deviations are provided in the ECEF (Earth-Centered, Earth-Fixed) coordinate system: sdx, sdy, and sdz represent the X, Y, and Z components in meters. Additionally, the values sdne, sdeu, sdun (or sdxy, sdyz, and sdzx in ECEF format) represent the square root of the absolute value of the corresponding covariance matrix components, with the sign indicating the direction of correlation. Using all these values, the user can reconstruct the full covariance matrix of the estimated position.
(6)	Age of differential (age)	The time difference between the observation data epochs of the target receiver and the base station in second.
(7)	Ratio factor (ratio)	The ratio factor of ″ratio-test″ for standard integer ambiguity validation strategy. The value means the ratio of the squared sum of the residuals with the second best integer vector to with the best integer vector.

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No	Record/Field	Description
(8)	Receiver Velocity*	The estimated velocity values of the solution are computed based on the positioning options. If out-solformat is set to 0:llh, the velocity is output in the ENU (East, North, Up) coordinate system: ve, vn, and vu represent the east, north, and up velocity components in meters per second. If out-solformat is set to 1:xyz, the velocity is output in the ECEF (Earth-Centered, Earth-Fixed) coordinate system: vx, vy, and vz represent the X, Y, and Z velocity components in meters per second. If out-solformat is set to 2:enu, velocity values are not output. These values allow users to understand the motion of the receiver in the respective coordinate system.
(9)	Velocity Standard deviations*	The estimated velocity values of the solution are computed based on the positioning options. If out-solformat is set to 0:llh, the velocity standard deviations are output in the ENU (East, North, Up) coordinate system: sdve, sdvn, and sdvu represent the east, north, and up components of the velocity standard deviations in meters per second. Additionally, sdvne, sdveu, and sdvun represent the square root of the absolute value of the NE, EU, and UN components of the estimated velocity covariance matrix, with the sign indicating the direction of correlation. If out-solformat is set to 1:xyz, the velocity standard deviations are output in the ECEF (Earth-Centered, Earth-Fixed) coordinate system: sdvx, sdvy, and sdvz represent the X, Y, and Z components of the velocity standard deviations in meters per second. Similarly, sdvxy, sdvyz, and sdvzx represent the square root of the absolute value of the XY, YZ, and ZX components of the velocity covariance matrix, with the sign indicating the direction of correlation. If out-solformat is set to 2:enu, velocity standard deviations are not output. These values allow users to reconstruct the full covariance matrix of the estimated velocity and assess the reliability of the motion solution.

* These items are output only when the Output Velocity option is enabled.

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B.2 Solution status file

DESCRIPTION

A solution status file is output of RTKNAVI or RTKPOST, that is a text file which contains the internal status of the positioning process. The internal status include estimated states of Kalman filter and residuals of measurements to analyze the solution quality. The following table shows the format of the solution status file when the positioning mode is set to PPP.

Table 5 Format of the solution status file

No	Record/Field	Description
1	Position States	Estimated target position in the filter. $POS,week,tow,stat,posx,posy,posz,possdx,possdy,possdz week/tow : gps week no/time of week (s) stat : solution status posx/posy/posz : position x/y/z ecef (m) possdx/possdy/possdz : standard deviations of position x/y/z ecef (m)
2	Velocity/ Acceleration States	Estimated target velocity and acceleration in the filter. $VELACC,week,tow,stat,vele,veln,velu,acce,accn,accu week/tow : gps week no/time of week (s) stat : solution status vele/veln/velu : velocity e/n/u (m/s) acce/accn/accu : acceleration e/n/u (m/s^2)
3	Velocity	Estimated target velocity in the filter. $VEL,week,tow,stat,velx,vely,velz,qvelx,qvely,qvelz week/tow : gps week no/time of week (s) stat : solution status velx/vely/velz : velocity ecef-x,y,z (m/s) qvelx/qvely/qvelz : velocity quality ecef-x,y,z (m/s)
4	Ionosphere Delay States	Estimated L1 ionosphere delay. $ION,week,tow,stat,sat,az,el,ion,ionsd week/tow : gps week no/time of week (s) stat : solution status sat : satellite id az/el : azimuth/elevation angle(deg) ion : ionospheric delay L1 (m) ionsd: standard deviations of ionospheric delay L1 (m)

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No	Record/Field	Description
5	Troposphere Delay States	Estimated zenith troposphere delay. $TROP,week,tow,stat,rcv,ztd,ztdsd week/tow : gps week no/time of week (s) stat : solution status rcv : receiver (1:target) ztd : zenith troposphere delay (m) ztdsd : standard deviations of zenith troposphere delay (m)
6	Troposphere Gradient States	Estimated zenith troposphere gradient. $TRPG,week,tow,stat,rcv,gradn,grade,gardnsd,gradesd week/tow : gps week no/time of week (s) stat : solution status rcv : receiver (1:target) gradn : tropospheric gradient to north (m) grade: tropospheric gradient to east (m) gradnsd : standard deviations to north (m) gradesd: standard deviations to east (m)
7	Attitude Data	$ATTE,week,tow,stat,attr,attp,atth,qattr,qattp,qatth week/tow : gps week no/time of week (s) stat : solution status attr,attp,atth : Attitude data roll/pitch/heading (deg) qattr,qattp,qatth : Attitude data quality roll/pitch/heading (deg)
8	Accelerometer Bias	$ACCB,week,tow,stat,accbx,accby,accbz,qaccbx,qaccby,qaccbz week/tow : gps week no/time of week (s) stat : solution status accbx/accby/accbz : Accelerometer bias x/y/z (m/s2) qaccbx/qaccby/qaccbz : Accelerometer bias quality x/y/z (m/s2)
9	Gyroscope Bias	$GYROB,week,tow,stat,gyrobr,gyrobp,gyrobh,qgyrobr,qgyrobp,qgyrobh week/tow : gps week no/time of week (s) stat : solution status gyrobr/gyrobp/gyrobh : gyroscope bias roll/pitch/heading (deg/s) qgyrobr/qgyrobp/qgyrobh :gyroscope bias quality roll/pitch/heading (deg/s)
10	Clock Drift	$CLKD,week,tow,stat,clkd,qclkd week/tow : gps week no/time of week (s) stat : solution status clkd : Clock drift(s/s) qclkd : Clock drift quality(s/s)

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No	Record/Field	Description
11	Receiver Clock - Bias States	Estimated receiver clock bias parameters. $CLK,week,tow,stat,rcv,clk1,clk2,clksd1,clksd2 week/tow : gps week no/time of week (s) stat : solution status rcv : receiver (1:target) clk1 : receiver clock bias GPS (ns) clk2 : receiver clock bias GLONASS (ns) clksd1 : standard deviations of receiver clock bias GPS (ns) clksd2 : standard deviations of receiver clock bias GLONASS (ns)
12	Inter-System Bias and Quality Indicators	Estimated receiver inter-system bias in the filter. $ISB,week,tow,stat,isb1,isb2,isb3,isb4,isb5,isb6,isb7,isb8, qisb1,qisb2,qisb3,qisb4,qisb5,qisb6,qisb7,qisb8 week/tow : gps week no/time of week (s) stat : solution status isb1-isb8 : inter-system bias for GLONASS / Galileo / QZSS/BeiDou-3/ BeiDou-2/BeiDou-2I/BeiDou-2G/BeiDou-2M (ns) qisb1–qisb8 : quality of inter-system bias for GLONASS / Galileo / QZSS/BeiDou-3/ BeiDou-2/BeiDou-2I/BeiDou-2G/BeiDou-2M (ns)
13	Dilution of Precision Indicators	Number of satellites and dilution of precision (DOP). $DOP,week,tow,stat,ns,gdop,pdop,hdop,vdop week/tow : gps week no/time of week (s) stat : solution status ns : number of satellites gdop/pdop/hdop/vdop : dilution of precision (GDOP / PDOP / HDOP / VDOP)
14	Substitute Positioning Status	Substitute positioning status. $SUB,week,tow,stat,sub week/tow : gps week no/time of week (s) stat : solution status sub : substitute positioning status (0: not applied, 1: broadcast ephemeris, 4: real-time ephemeris, 9: autonomous navigation)
15	Ambiguity Parameter States	Estimated ambiguity parameter. $AMB,week,tow,stat,sat,amb,ambsd week/tow : gps week no/time of week (s) stat : solution status sat/frq : satellite id/frequency (1:L1,2:L2,3:L5,…) amb : ambiguity (m) ambsd : standard deviations of ambiguity (m)

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B.3 Configuration file for RNX2RTKP and RTKRCV

DESCRIPTION

A configuration file containing processing options, solution options and file options for RNX2RTKP and RTKRCV. That is a text file which contains the Keyword = Value form records indicating the various options. For enumeration values, the selectable value is either of a number (0,1,2,…) or an enumeration label (off, on, …). The line starting with # and the texts after # in a line are treated as comments.

Table 6 Common configuration file format for RNX2RTKP and RTKRCV

No.	Item	Configuration File	Descriptions	Notes
1	Positioning Mode	pos1-posmode	Defines the positioning mode (0:single,1:dgps,2:kinematic,3:static,4:moving base,5:fixed,6:ppp-kine,7:ppp-static,8:ppp-fixed) -Single : Single point positioning or SBAS DGPS -DGPS/DGNSS : Code-based differential GPS -Static : Carrier-based Static positioning -Kinematic: Carrier-based Kinematic positioning -Moving-Base: Moving baseline -Fixed: Target receiver position is fixed * -PPP Kinematic: Precise Point Positioning with kinematic mode -PPP Static: Precise Point Positioning with static mode -PPP Fixed: Target receiver position is fixed with PPP mode *	* For residuals analysis
2	Frequencies	pos1-frequency	Defines the used carrier frequencies (1:l1,2:l1+2,3:l1+2+3) -L1 : L1 Single frequency -L1+2 : L1 and L2 Dual-frequency -L1+2+5: L1, L2 and L5 Triple-frequency	
3	Filter Type	pos1-soltype	Defines the filter type (0:forward,1:backward,2:combined) -Forward: Forward filter solution -Backward: Backward filter solution * -Combined: Smoother combined solution with forward and backward filter solutions *	* N/A to RTKRCV and Single mode
4	Elevation Mask	pos1-elmask	Defines the elevation mask angle in degree.	

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No.	Item	Configuration File	Descriptions	Notes
5	SNR Mask : Target	pos1-snrmask_r	Defines the whether to enable the SNR mask for the target receiver. (0:off,1:on)	
6	SNR Mask : Base-Station	pos1-snrmask_b	Defines the whether to enable the SNR mask for the base station. (0:off,1:on)	
7	SNR Mask : Threshold L1	pos1-snrmask_L1	Defines the SNR thresholds used to reject L1 satellite signals based on elevation angle. Specify the thresholds as comma-separated values, where each value corresponds to a 5-degree elevation bin (e.g., 0–5°, 5–10°, …, 85–90°).	
8	SNR Mask : Threshold L2	pos1-snrmask_L2	Defines the SNR thresholds used to reject L2 satellite signals based on elevation angle.	
9	SNR Mask : Threshold L5	pos1-snrmask_L5	Defines the SNR thresholds used to reject L5 satellite signals based on elevation angle.	
10	Receiver Dynamics	pos1-dynamics	Defines the dynamics model of the target receiver. (0:off,1:dynamics,2:ins) -OFF: Dynamics is not used -Dynamics: Receiver velocity and acceleration are estimated. The receiver position is predicted with the estimated velocity and acceleration. -Ins: Performs PPP positioning using INS-based inertial navigation.	Only applicable to DGPS/DGNSS or Kinematic modes
11	Substitute PPP Option	pos1-subopt	Defines the substitute PPP mode selection. (0:off,1:brdc, 4:rt) -OFF : Not applied -BRDC : Uses broadcast ephemeris -RT : Uses real-time ephemeris	
12	Dead Reckoning Interval	pos1-drint	Defines the dead reckoning interval in seconds. If a value greater than 0 is Defines the, this setting will be used.	

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No.	Item	Configuration File	Descriptions	Notes
13	Earth Tides Correction	pos1-tidecorr	Defines the whether earth tides correction is applied or not (0:off,1:on,2:otl) -OFF: Not apply earth tides correction -Solid: Apply solid earth tides correction -Solid/OTL: Apply solid earth tides, OTL (ocean tide loading) and pole tide corrections. * To apply OTL correction, set the OTL coefficients file path using “file-blqfile” in Ocean Loading BLQ format. The marker name must be included in the input RINEX file to match the station in the BLQ file. To apply pole tide correction, set the ERP file path using “file-eopfile” in EOP format.	N/A to Single mode *N/A to RTKRCV
14	Ionosphere Correction	pos1-ionoopt	Defines the ionospheric correction options. If you select the "Estimate" parameter, the ionospheric delay for each satellite is estimated. (0:off, 1:brdc, 2:sbas, 3:dual-freq, 4:est-stec, 5:ionex-tec, 6:qzs-brdc) OFF : Do not apply ionospheric correction Broadcast : Apply broadcast ionospheric model SBAS : Apply SBAS ionospheric model Iono-Free LC : Apply ionosphere-free linear combination using the selected dual-frequency signals Estimate iono: Estimate the ionospheric delay parameter for each satellite IONEX TEC : Use IONEX TEC grid data QZSS Broadcast : Apply broadcast ionosphere model provided by QZSS	
15	Troposphere Correction	pos1-tropopt	Defines the whether tropospheric parameters (zenith total delay at target and base-station positions) are estimated or not. (0:off,1:saas,2:sbas,3:est-ztd,4:est-ztdgrad) -OFF : Not apply troposphere correction -Saastamoinen: Apply Saastamoinen model -SBAS: Apply SBAS tropospheric model (MOPS) -Estimate ZTD: Estimate ZTD (zenith total delay) parameters as EKF states * -Estimate ZTD+Grad: Estimate ZTD and horizontal gradient parameters as EKF states *	* N/A to Single mode

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No.	Item	Configuration File	Descriptions	Notes
16	Satellite Ephemeris/ Clock	pos1-sateph	Defines the type of satellite ephemeris. (0:brdc,1:precise,2:brdc+sbas,3:brdc+ssrapc,4:brdc+ssrcom) -Broadcast : Use broadcast ephemeris -Precise : Use precise ephemeris * -Broadcast+SBAS: Broadcast ephemeris with SBAS long-term and fast correction -Broadcast+SSR APC: Broadcast ephemeris with RTCM SSR correction (antenna phase center value) -Broadcast+SSR CoM: Broadcast ephemeris with RTCM SSR correction (satellite center of mass value)	
17	Satellite PCV	pos1-posopt1	Defines the whether the satellite antenna PCV (phase center variation) model is used or not. To use the feature, Defines the ″file-satantfile″. (0:off,1:on)	N/A to Single mode
18	Receiver PCV	pos1-posopt2	Defines the whether the receiver antenna PCV model is used or not. To use the feature, Defines the ″file-rcvantfile″. (0:off,1:on)	N/A to Single mode
19	Phase Windup	pos1-posopt3	Defines the whether the phase windup correction for PPP modes is applied or not. (0:off,1:on,2:precise)	Only applicable to PPP-* modes.
20	Reject Eclipse	pos1-posopt4	Defines the whether the GPS Block IIA satellites in eclipse are excluded or not. The eclipsing Block IIA satellites often degrade the PPP solutions due to unpredicted behavior of yaw-attitude. (0:off,1:on)	Only applicable to PPP-* modes.
21	RAIM FDE	pos1-posopt5	Defines the whether RAIM (receiver autonomous integrity monitoring) FDE (fault detection and exclusion) feature is enabled or not. In case of RAIM FDE enabled, a satellite is excluded if SSE (sum of squared errors) of residuals is over a threshold. The excluded satellite is selected to indicate the minimum SSE. (0:off,1:on)	
22	Day-Boundary Jump	pos1-posopt6	Enable day-boundary clock jump detection and correction. When enabled, the system checks if the observation time is exactly at the GPS day boundary (00:00:00), which may cause a discontinuity in the receiver clock. This helps mitigate positioning errors due to sudden time jumps at midnight. (0:off,1:on)	Only applicable to PPP-* modes.

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No.	Item	Configuration File	Descriptions	Notes
23	Excluded Satellites (+PRN: Included)	pos1-exclsats	Defines the excluded satellites for positioning. Fill in the PRN numbers of the satellites separated by spaces. For GLONASS, Galileo, QZSS, BeiDou and SBAS, use Rnn, Enn, Jnn, Cnn and Snn, respectively (nn: satellite PRN or slot number). If "+" is added to the head of the satellite ID, the satellite is included for positioning even if the satellite is unhealthy.	
24	Navigation System	pos1-navsys	Set the satellite systems to be used for positioning using a bit value. If a system's corresponding bit is set to 0, its satellites will not be used for positioning. (Bit Array : 1:gps+2:sbas+4:glo+8:gal+16:qzs+32:bds+64:navic)	
25	SNR Weighting Threshold	pos1-snrwthres_gps	Specify the value in the format "XX = threshold." XX represents the signal identifier in RINEX format (numerical value + alphabet). * Defines the SNR weighting threshold. Used to determine the validity of satellite signals based on elevation angle. The threshold values correspond to elevation angle intervals of 5° (ex., 0–5°, 10°, 15° … 85–90°) and should be specified as comma-separated values. Configure individually for each satellite system and signal type.	*The available identifiers are shown in Table 8.
26		pos1-snrwthres_glo		
27		pos1-snrwthres_gal		
28		pos1-snrwthres_qzs		
29		pos1-snrwthres_bds		
30	SNR Weighting : Margin	pos1-snrwmargin_gps	Defines the SNR weighting margin. The SNR weighting margin represents the valid range (in dB) for satellite signals whose SNR falls below the SNR weighting threshold. Configure individually for each satellite system.	
31		pos1-snrwmargin_glo		
32		pos1-snrwmargin_gal		
33		pos1-snrwmargin_qzs		
34		pos1-snrwmargin_bds		
35	SNR Weighting : Error Ratio	pos1-snrweratio_gps	Defines the SNR weighting error standard deviation increase rate. defines the rate at which the observation error standard deviation increases per dB within the range below the SNR weighting threshold	
36		pos1-snrweratio_glo		

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No.	Item	Configuration File	Descriptions	Notes
37		pos1-snrweratio_gal	(i.e., within the SNR weighting margin). For example: • If the threshold is 45 dB, the margin is 10 dB, and the increase rate is 10, • Then a signal with 44 dB SNR will have an error multiplier of 10 applied.	
38		pos1-snrweratio_qzs		
39		pos1-snrweratio_bds		
40	MADOCA-PPP Ionospheric Correction	pos2-ionocorr	Defines whether to use wide-area ionospheric correction information.(0:off,1:on)	
41	Integer Ambiguity Resolution	pos2-armode	Defines the strategy of integer ambiguity resolution (0:off,1:continuous,2:instantaneous,3:fix-and-hold) -OFF : No ambiguity resolution -Continuous : Continuously static integer ambiguities are estimated and resolved * -Instantaneous : Integer ambiguity is estimated and resolved by epoch-by-epoch basis * -Fix and Hold : Continuously static integer ambiguities are estimated and resolved. If the validation OK, the ambiguities are tightly constrained to the resolved values. *	Default: Continuous Not applicable to Single mode. * Only applicable to Kinematic, Static, Movingbase line , Fixed and PPP-* modes.
42	GNSS Used for PPP-AR	pos2-arsys	Set the satellite systems to be used for integer ambiguity resolution using a bit value. (Bit Array: 1:gps+8:gal+16:qzs+32:bds)	
43	Cascading Ambiguity Resolution	pos2-car	CAR (Cascading Ambiguity Resolution) enable flag. (0:off,1:on)	
44	Min Ratio to Fix Ambiguity	pos2-arthres	Defines the integer ambiguity validation threshold for ″ratio-test″, which uses the ratio of squared residuals of the best integer vector to the second-best vector.	Default value:3.0
45	Min Confidence to Fix Amb.	pos2-arthres1	Defines the minimum confidence level to fix ambiguity in PPP-AR mode	
46	Max FCB to Fix Amb.	pos2-arthres2	Defines the maximum FCB (fractional cycle bias) to fix ambiguity in PPP-AR mode	
47	Max WL Deviation to Fix Ambiguity	pos2-arthres3	Defines the maximum deviation between the float and rounded wide-lane ambiguity to fix ambiguity in PPP-AR mode.	
48	Max NL Deviation to Fix Ambiguity	pos2-arthres4	Defines the maximum deviation between the float and rounded narrow-lane ambiguity to fix ambiguity in PPP-AR mode.	

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No.	Item	Configuration File	Descriptions	Notes
49	CAR Threshold1	pos2-carthres1	Validation ratio for lambda search with CAR.	
50	CAR Threshold2	pos2-carthres2	Maximum 3D-position std-dev to start narrow-lane ambiguity search with CAR. (m)	
51	Min Lock to Fix Ambiguity	pos2-arlockcnt	Defines the minimum lock count to fix integer ambiguity. If the lock count or the elevation angle is less than the value, the ambiguity is excluded from the fixed integer vector.	Default value:0
52	Min Elevation to Fix Ambiguity	pos2-arelmask	Defines the minimum elevation angle (deg) to fix integer ambiguity. If the lock count or the elevation angle is less than the value, the ambiguity is excluded from the fixed integer vector.	Default value:0
53	Min Fix to Hold Ambiguity	pos2-arminfix	If you select ″Fix and Hold″ mode for Integer Ambiguity Resolution (pos2-armode), set the minimum fix count to hold ambiguity.	Default value:10
54	Max AR Iterations	pos2-armaxiter	Defines the maximum iterations for fixing integer ambiguities. If validation fails within this limit, fixing is aborted.	
55	Min Elevation to Hold Ambiguity	pos2-elmaskhold	If you select ″Fix and Hold″ mode for Integer Ambiguity Resolution (pos2-armode), set the minimum elevation angle (deg) to hold ambiguity.	
56	Outage to Reset Ambiguity	pos2-aroutcnt	Defines the outage count to reset ambiguity. If the data outage count is over the value, the estimated ambiguity is reset to the initial value.	
57	Max Age of Differential	pos2-maxage	Defines the maximum value of age of differential (s) between the target and the base station.	
58	Outage to Slip Threshold for GF	pos2-slipthres	Defines the cycle-slip threshold (m) of Geometry-Free LC carrier-phase difference between epochs.	
59	Reject Threshold of Innovation	pos2-rejionno	Defines the reject threshold of innovation (pre-fit residual) (m). If the innovation is over the value, the observable is excluded for the estimation process as an outlier.	
60	Reject Threshold of GDOP	pos2-rejgdop	Defines the reject threshold of GDOP. If the GDOP is over the value, the observable is excluded for the estimation process as an outlier.	
61	Number of Iteration	pos2-niter	Defines the number of iteration in the measurement update of the estimation filter. If the baseline length is very short like 1 m, the iteration may be effective to handle the nonlinearity of measurement equation.	Default value: 1

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No.	Item	Configuration File	Descriptions	Notes
62	Baseline Length	pos2-baselen	If moving-base mode is enabled, set the baseline length constraint (m).	
63	Baseline Length standard Deviation	pos2-basesig	If moving-base mode is enabled, set the standard deviation (m) of the baseline length constraint.	
64	Signal Selection Option GPS IIR-M	pos2-siggpsIIR-M	Defines the signal selection for GPS BLOCK IIR-M. (0:L1C/A-L2P,1:L1C/A-L2C ,3:L1C/A-L2)	
65	Signal Selection Option GPS IIF	pos2-siggpsIIF	Defines the signal selection for GPS BLOCK IIF. (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)	
66	Signal Selection Option GPS IIIA	pos2-siggpsIIIA	Defines the signal selection for GPS BLOCK IIIA. (0:L1C/A-L2P,1:L1C/A-L2C,2:L1C/A-L5,3:L1C/A-L2,4:L1C/A-L2P-L5,5:L1C/A-L2C-L5)	
67	Signal Selection Option for GAL	pos2-siggal	Defines the signal selection option for Galileo. (0:E1-E5a,1:E1-E5b,2:E1-E5a-E5b,3:E1-E5a-E5b-E6)	
68	Signal Selection Option QZS1/2	pos2-sigqzs1_2	Defines the signal selection for QZSS Block I and II (QZS02,03,04). (0:L1C-L5,1:L1C/A-L2C ,3:L1C-L2C-L5,4:L1C/A-L2C-L5)	
69	Signal Selection Option QZS-2A	pos2-sigqzs2A	Defines the signal selection QZSS Block IIA (QZS1R). (0:L1C-L5,1:L1C/A-L2C,2:L1C/B-L5,3:L1C-L2C-L5,4:L1C/A-L2C-L5,5:L1C/B-L2C-L5)	
70	Signal Selection Option QZS-3	pos2-sigqzs3	Defines the signal selection QZSS Block III (QZS05,06,07). (0:L1C-L5 ,2:L1C/B-L5)	
71	Signal Selection Option BDS-2	pos2-sigbds2	Defines the signal selection BeiDou-2. (0:B1I-B3I,1:B1I-B2I,2:B1I-B3I-B2I)	
72	Signal Selection Option BDS-3	pos2-sigbds3	Defines the signal selection BeiDou-3. (0:B1-B3,1:B1C-B2a,2:B1I-B3I-B2a)	
73	Chi-Squared Test Anomaly Disable Option	pos2-ign_chierr	Ignore chi-square error mode (0:off,1:on) When it's activated, it ignores chi-square error and continues single point positioning.	

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No.	Item	Configuration File	Descriptions	Notes
74	BDS-2 ISB Estimation	pos2-bds2bias	Estimates the inter-system bias (ISB) for BeiDou-2.(0:off,1:on)	
75	Satellite Code Bias for PPP	pos2-pppsatcb	Defines the method used for satellite code bias correction. (0:auto,1:l6e,2:bia,3:dcb,4:rtcm3) -Auto : Selects the first available valid bias correction method in the order of: l6e, bia, dcb, rtcm3. -L6E : Uses L6E signal-based bias correction. -BIA : Applies BIA (Bias Information Analysis) method. -DCB : Uses Differential Code Bias data. -RTCM3 : Applies RTCM Version 3 standard bias correction.	
76	Satellite Phase Bias for PPP	pos2-pppsatpb	Defines the method for selecting satellite phase bias. (0:auto,1:l6e,2:bia,3:fcb,4:rtcm3)	
77	Uncorrected Code Bias	pos2-uncorrbias	0:use uncorrected code bias, 1: do not use uncorrected code bias. (0:use,1:notuse,2:notuse-uncorrsatbias)	
78	Bias Validity Time Threshold	pos2-maxbiasdt	Defines the Bias SINEX code bias timeout in seconds.	
79	Missing number	-	-	
80	Processing Interval	pos2-tint	Defines the parameter used to relax the threshold when detecting and removing outliers based on the standard deviation of PPP residual observation errors, taking into account missing data periods. (s)	
81	Clock Estimation	pos2-clkopt	Estimation option of receiver clock bias with random walk.(0:off,1:on) If pos1-dynamics is set to ins, this setting is used.	
82	Clock Drift Estimation	pos2-clkdopt	Estimation option of receiver clock drift with random walk. (0:off,1:on) If pos1-dynamics is set to ins, this setting is used.	
83	IMU Measurement Interval	pos2-insint	Defines the measurement interval for IMU.(s) This value is used to determine the update interval and compute the process noise for INS inertial navigation.	

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No.	Item	Configuration File	Descriptions	Notes
84	INS Data Outage Detection Threshold	pos2-insreset	Defines the timeout threshold for INS sensor data. When the outage duration exceeds this value, the position state is initialized with the stand-alone solution. (s)	
85	Orbit Dynamics Model Initialization Threshold	pos2-maxdist	Defines the initialization threshold for the orbit dynamics model. If the distance between the current position solution and the model exceeds this value, the model is reinitialized. (m)	
86	Observation-Excluded Object	pos2-maskobj	Defines the objects to be excluded from observation in the azimuth/elevation mask definition. For example, "STVX" excludes all objects. (S:Structure,T:Terrain,V:Vegetation,X:eXtra)	
87	Solution Format	out-solformat	Defines the output solution format. (0:llh,1:xyz,2:enu,3:nmea) -Lat/Lon/Height : Latitude, longitude and height -X/Y/Z-ECEF : X/Y/Z components of ECEF coordinates -E/N/U-Baseline: E/N/U components of baseline vector -NMEA0183 : NMEA GPRMC, GPGGA, GPGSA, GLGSA, GAGSA, GPGSV, GLGSV and GAGSV	For RTKRCV, specify options as Output Streams setting.
88	Output Header	out-outhead	Defines the whether the header is output or not. (0:off,1:on)	N/A to NMEA
89	Output Processing Options	out-outopt	Defines the whether the processing options are output or not. (0:off,1:on)	RNX2RTKP only N/A to NMEA
90	Output Velocity	out-outvel	Defines the whether the velocity is output or not. (0:off, 1:on)	N/A to NMEA
91	Time System	out-timesys	Defines the time system. (0:gpst,1:utc,2:jst) -GPST: GPS Time, a continuous time scale without leap seconds. -UTC: Coordinated Universal Time, includes leap seconds. -JST: Japan Standard Time (UTC+9).	N/A to NMEA
92	Time Format	out-timeform	Defines the format of time(0:tow,1:hms) -Tow: GPS week and time of week (ssssssss.sss GPST). -HMS: Date and time in the format yyyy/mm/dd hh:mm:ss, displayed in GPST, UTC, or JST.	N/A to NMEA

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No.	Item	Configuration File	Descriptions	Notes
93	# of Decimals	out-timendec	Defines the number of decimals in the time format.	N/A to NMEA
94	Latitude/Longitude Format	out-degform	Defines the formats of latitude and longitude if the solution format is set to Lat/Lon/Height.(0:deg,1:dms) -Deg : Degree (ddd.dddddddd) -DMS : Degree minute second (ddd mm ss.sss)	N/A to NMEA
95	Field Separator	out-fieldsep	Defines the separator for fields.	N/A to NMEA
96	Single Solution Output	out-outsingle	Defines whether the single point positioning result is output when the positioning mode is not set to single. (0:off, 1:on) -OFF: Suppresses the output of single point positioning results. -ON: Outputs single point positioning results even when using relative or precise positioning modes.	
97	Maximum Solution Standard Deviation	out-maxsolstd	Defines the threshold for the standard deviation of the positioning solution. If the solution's standard deviation exceeds this value, the output is suppressed. A value of zero disables this filtering. (m)	
98	Type of Height	out-height	Defines the type of height. (0:ellipsoidal,1:geodetic) -Ellipsoidal : Ellipsoidal height -Geodetic : Geodetic height	
99	Geoid Model	out-geoid	Defines the geoid model if the Height option is set to Geodetic. (0:internal,1:egm96,2:egm08_2.5,3:egm08_1, 4:gsi2000) -Internal: Internal geoid model -EGM96-BE (15″) : EGM96 (15″ x 15″ grid) *1 -EGM2008-SE (2.5″) : EGM2008 (2.5 x 2.5″ grid) *2 -EGM2008-SE (1″): EGM2008 (1 x 1″ grid) *2 -GSI2000 (1x1.5″): GSI2000 (1x1.5″ grid) *3 If using external geoid model, specify the geoid file path in ″file-geoidfile”.	

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No.	Item	Configuration File	Descriptions	Notes
100	Solution for Static Mode	out-solstatic	Defines the solution type for Static or PPP-Static mode (0:all,1:single) -All: all solutions for the processing period are outputted -Single: Only one solution for the processing period is output. The time of solution is first epoch in the processing period.	RNX2RTKP only
101	NMEA Interval (s) RMC/GGA	out-nmeaintv1	Defines the output interval of NMEA GPRMC, GPGGA messages	RTKRCV only
102	NMEA Interval (s) GSA/GSV	out-nmeaintv2	Defines the output interval of NMEA GPGSA, GLGSA, GAGSA, GPGSV, GLGSV, GAGSV messages	RTKRCV only
103	Output Solution Status	out-outstat	Defines the output level of the solution status file. The solution status file contains estimated states and residuals. (0:off,1:state,2:residual) The solution status file is created in the current directory (RTKRCV) or in the output file directory (RNX2RTKP).	
104	Code/Carrier-Phase Error Rate L1	stats-eratio1	Defines the ratio of standard deviations of pseudorange errors to carrier-phase errors for L1 , used in both standalone and RTK positioning.	Default value: 100
105	Code/Carrier-Phase Error Rate L2	stats-eratio2	Defines the ratio of standard deviations of pseudorange errors to carrier-phase errors for L2/L5/L6 , used in both standalone and RTK positioning.	Default value: 100
106	Carrier-Phase Error	stats-errphase	Defines the base term of carrier-phase error standard deviation , used in both standalone and RTK positioning (m).	Default value: 0.003
107	Carrier-Phase Error/sinEl	stats-errphaseel	Defines the elevation dependent term of carrier-phase error standard deviation , used in both standalone and RTK positioning (m/sin(el)).	Default value: 0.003
108	Carrier-Phase Error/Baseline	stats-errphasebl	Defines the baseline-length dependent term of carrier-phase error standard deviation (m/10km).	Default value: 0
109	Doppler Frequency	stats-errdoppler	Defines the standard deviation of Doppler errors (Hz) (Current version does not use the value)	Default value: 1
110	URA Ratio	stats-uraratio	Defines the URA(User Range Accuracy) ratio.	

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No.	Item	Configuration File	Descriptions	Notes
111	Initial Standard Deviation : Carrier-Phase Bias	stats-stdbias	Defines the initial standard deviation of the carrier-phase bias.	Only applicable to Static, Kinematic, Moving or Fixed modes
112	Initial Standard Deviation : Ionospheric Delay	stats-stdiono	Defines the initial standard deviation of the ionospheric delay. If Ionosphere Correction is set to Estimate STEC , this setting is used.	Only applicable to Static, Kinematic, Moving or Fixed modes
113	Initial Standard Deviation : Tropospheric Delay	stats-stdtrop	Defines the initial standard deviation of the tropospheric delay. If Troposphere Correction is set to Estimate ZTD/ZTD+Grad , this setting is used.	Only applicable to Static, Kinematic, Moving or Fixed modes
114	Initial Standard Deviation : Receiver Attitude	stats-stdatt	Defines the initial standard deviation of the receiver attitude. (deg)	
115	Initial Standard Deviation : Receiver Velocity	stats-stdvel	Defines the initial standard deviation of the receiver velocity. (m/s)	
116	Initial Standard Deviation : Receiver Position	stats-stdpos	Defines the initial standard deviation of the receiver position. (m)	
117	Initial Standard Deviation : Receiver Accelerometer Bias	stats-stdaccb	Defines the initial standard deviation of the receiver accelerometer bias. (micro-g)	
118	Initial Standard Deviation : Receiver Gyro Bias	stats-stdgyrob	Defines the initial standard deviation of the receiver gyro bias. (deg/h)	

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No.	Item	Configuration File	Descriptions	Notes
119	Initial Standard Deviation : Receiver Clock Offset	stats-stdclko	Defines the initial standard deviation of the receiver clock offset. (m)	
120	Initial Standard Deviation : Receiver Clock Drift	stats-stdclkd	Defines the initial standard deviation of the receiver clock drift. (m/s)	
121	Process Noise : Receiver Accel Horizontal	stats-prnaccelh	Defines the process noise standard deviation of the receiver acceleration as the horizontal component. (m/s2/sqrt(s)). If Receiver Dynamics is set to ON , this setting is used.	Default value: 1
122	Process Noise : Receiver Accel Vertical	stats-prnaccelv	Defines the process noise standard deviation of the receiver acceleration as the vertical component. (m/s2/sqrt(s)). If Receiver Dynamics is set to ON , this setting is used.	Default value: 0.1
123	Process Noise : Carrier-Phase Bias	stats-prnbias	Defines the process noise standard deviation of carrier-phase bias (ambiguity) (cycle/sqrt(s)).	Default value: 1E-4
124	Process Noise : Ionospheric Delay	stats-prniono	Defines the process noise standard deviation of the ionospheric delay (m/sqrt(s)).	Default value: 1E-3
125	Process Noise : Zenith Tropospheric Delay	stats-prntrop	Defines the process noise standard deviation of zenith tropospheric delay (m/sqrt(s)).	Default value: 1E-4
126	Process Noise : Static Position	stats-prnpos	Defines the process noise standard deviation of static position (m/sqrt(s)). If Positioning Mode is set to Static or PPP-Static, this setting is used.	
127	Process Noise : Inter-Frequency Bias	stats-prnifb	Defines the process noise for the inter-frequency bias of the third and fourth frequencies. (m)	
128	Process Noise : Receiver Clock Drift	stats-prnclkd	Defines the process noise deviation of the receiver clock drift. (m)	

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No.	Item	Configuration File	Descriptions	Notes
129	Process Noise : Receiver GPS Clock Offset	stats-prnclk_gps	Defines the process noise deviation of the receiver GPS clock offset. (m)	
130	Process Noise : Receiver GLO Clock Offset	stats-prnclk_glo	Defines the process noise deviation of the receiver GLONASS clock offset. (m)	
131	Process Noise : Receiver GAL Clock Offset	stats-prnclk_gal	Defines the process noise deviation of the receiver Galileo clock offset. (m)	
132	Process Noise : Receiver QZS Clock Offset	stats-prnclk_qzs	Defines the process noise deviation of the receiver QZSS clock offset. (m)	
133	Process Noise : Receiver BDS Clock Offset	stats-prnclk_bds	Defines the process noise deviation of the receiver BeiDou clock offset. (m)	
134	Process Noise : Receiver IRN Clock Offset	stats-prnclk_irn	Defines the process noise deviation of the receiver IRNSS clock offset. (m)	
135	Process Noise : Receiver BDS-2 Clock Offset	stats-prnclk_bds2	Defines the process noise for the BeiDou-2 receiver clock offset. (m)	
136	Process Noise : Receiver Accelerometer Noise	stats-prnaccn	Defines the process noise deviation of the receiver accelerometer noise. (micro-g/Hz)	
137	Process Noise : Receiver Gyro Noise	stats-prngyron	Defines the process noise deviation of the receiver gyro noise. (deg/h)	
138	Process Noise : Receiver Accelerometer Bias	stats-prnaccb	Defines the process noise deviation of the receiver accelerometer bias. (micro-g/Hz)	

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No.	Item	Configuration File	Descriptions	Notes
139	Process Noise : Receiver Gyro Bias	stats-prngyrob	Defines the process noise deviation of the receiver gyro bias. (deg/h)	
140	Detecting Clock Jumps Threshold	stats-clkjump	Defines the threshold for detecting clock jumps(s)	
141	Satellite Clock Stability	stats-clkstab	Defines the satellite clock stability (s/s). The value is used for interpolation of base-station observables.	
142	Antenna1 : Position Type	ant1-postype	Defines the position type of the target antenna if the target antenna is fixed. (0:llh,1:xyz,2:single,3:posfile,4:rinexhead,5:rtcm,6:raw) -Lat/Lon/Height (deg/m): Latitude/longitude/height in degree and m -Lat/Lon/Height (dms/m): Latitude/longitude/height in degree/minute/second and m -X/Y/Z-ECEF (m): X/Y/Z components in ECEF frame. -RTCM Station Position: Use the antenna position included in RTCM messages * -Average of Single-Pos: Use the average of single point solutions ** -Get from Position File: Use the position in the position file. The station is searched by using the head 4-character ID of the target observation data file path. ** -RINEX Header Position: Use the approximate position in RINEX OBS header. **	Only applicable to Fixed or PPP Fixed modes. Height is specified as ellipsoidal height * RTKRCV only ** RNX2RTKP only
143	Antenna1 : Position 1	ant1-pos1	Defines the position 1 of the target antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (deg|m)	Only applicable to Fixed or PPP Fixed modes.
144	Antenna1 : Position 2	ant1-pos2	Defines the position 2 of the target antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (deg|m)	Only applicable to Fixed or PPP Fixed modes.
145	Antenna1 : Position 3	ant1-pos3	Defines the position 3 of the target antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (m|m)	Only applicable to Fixed or PPP Fixed modes.

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No.	Item	Configuration File	Descriptions	Notes
146	Antenna1 : Antenna Type	ant1-anttype	Defines the type of the target antenna. To select the antenna type, set the Receiver Antenna PCV File path in "file-rcvantfile". If "*" is used, antenna type and antenna delta are recognized by the antenna information of RINEX OBS header (RNX2RTKP) or RTCM antenna information (RTKRCV).	
147	Antenna1 : Delta-E	ant1-antdele	Defines the delta position of the target antenna as the east offsets of ARP (antenna reference point) position with refer to the marker (m).	
148	Antenna1 : Delta-N	ant1-antdeln	Defines the delta position of the target antenna as the north offsets of ARP (antenna reference point) position with refer to the marker (m).	
149	Antenna1 : Delta-U	ant1-antdelu	Defines the delta position of the target antenna as the up offsets of ARP (antenna reference point) position with refer to the marker (m).	
150	Lever Arm Between GNSS and IMU	ant1-antimux	Defines the lever arm between the GNSS and IMU body-x/y/z. (m)	
151		ant1-antimuy		
152		ant1-antimuz		
153	IMU Rotation	ant1-rotimux	Defines the rotation of the IMU body-x/y/z. (deg)	
154		ant1-rotimuy		
155		ant1-rotimuz		
156	Antenna2 : Position Type	ant2-postype	Defines the position type of the target antenna if the base-station antenna. (0:llh,1:xyz,2:single,3:posfile,4:rinexhead,5:rtcm,6:raw)	Only applicable to Static, Kinematic or Fixed modes.
157	Antenna2 : Position 1	ant2-pos1	Defines the position 1 of the base-station antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (deg|m)	Only applicable to Static, Kinematic or Fixed modes.
158	Antenna2 : Position 2	ant2-pos2	Defines the position 2 of the base-station antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (deg|m)	Only applicable to Static, Kinematic or Fixed modes.

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No.	Item	Configuration File	Descriptions	Notes
159	Antenna2 : Position 3	ant2-pos3	Defines the position 3 of the base-station antenna if the target antenna is fixed. Specify the value according to the Position Type setting. (m|m)	Only applicable to Static, Kinematic or Fixed modes.
160	Antenna2 : Antenna Type	ant2-anttype	Defines the type of the base-station antenna. To select the antenna type, set the Receiver Antenna PCV File path in "file-rcvantfile". If "*" is used, antenna type and antenna delta are recognized by the antenna information of RINEX OBS header (RNX2RTKP) or RTCM antenna information (RTKRCV).	
161	Antenna2 : Delta-E	ant2-antdele	Defines the delta position of the base-station antenna as the east offsets of ARP (antenna reference point) position with refer to the marker (m).	
162	Antenna2 : Delta-N	ant2-antdeln	Defines the delta position of the base-station antenna as the north offsets of ARP (antenna reference point) position with refer to the marker (m).	
163	Antenna2 : Delta-U	ant2-antdelu	Defines the delta position of the base-station antenna as the up offsets of ARP (antenna reference point) position with refer to the marker (m).	
164	Max Average Epochs	ant2-maxaveep	Defines the maximum number of epochs used to average the base station position in Single mode.	RTKRCV only
165	Init Reset	ant2-initrst	Defines whether to reset and reinitialize base position averaging on server start. (0:off,1:on)	
166	Time Interpolation of Base Station Observation Data	misc-timeinterp	Defines the ON to enable time interpolation of base station data. (0:off,1:on) If setting ON, the base station data are linearly interpolated to the target epoch and DD (double difference) is made with them. If setting off, nearest epoch of base station data is used for DD.	
167	SBAS Satellite Selection	misc-sbasatsel	If SBAS DGPS correction enabled, input SBAS satellite PRN number to be used. If you input 0, all available SBAS satellites are used.	

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No.	Item	Configuration File	Descriptions	Notes
168	RINEX Option (Target)	misc-rnxopt1	Specify RINEX read options for target RINEX observation data as follows. Multiple options can be input separated by spaces. -GLss[=+n.nn]: select GPS signal ss -RLss[=+n.nn]: select GLO signal ss -ELss[=+n.nn]: select GAL signal ss -JLss[=+n.nn]: select QZS signal ss -CLss[=+n.nn]: select BDS signal ss -SLss[=+n.nn]: select SBS signal ss ss : signal id =+n.nn: phase shift +n.nn (cycle) to be add to carrier-phase observables Without the option and with multiple signals in a frequency, a signal is selected for that frequency according to the default priorities of signals.	
169	RINEX Option (Base)	misc-rnxopt2	Specify RINEX read options for base-station RINEX observation data same as the RINEX Option (Target).	
170	PPP Option	misc-pppopt	Specify PPP options as follows. -GAP_RESION={number of epoch} : Sets the threshold (in epochs) to reset ionospheric parameters when observation gaps occur. -TRACE_AR : Enables logging of ambiguity resolution details during PPP processing.	
171	RTCM Option	misc-rtcmopt	Specify RTCM3 options for MADOCA-SSR as follows. -IGNORE_MCSSR_ALERT : Ignore the MADOCA-SSR Alert Flag -MCSSR_PRN={PRN} : Only decode MADOCA-SSR for the specified PRN -DIS_MCSSR_CB : Do not use madoca-ssr Code Bias	
172	Satellite Antenna PCV File ANTEX	file-satantfile	If you use the precise ephemeris or SSR correction, input the ANTEX antenna parameters file path for the satellite antenna PCV (phase center variation) correction. Usually use latest .atx file provided by IGS. For MADOCA-PPP (IS-QZSS-MDC-004), this setting is not required.	
173	Receiver Antenna PCV File ANTEX or NGS PCV	file-rcvantfile	If you apply the receiver antenna phase center offset and PCV correction, input ANTEX or NGS type antenna parameters file path.	

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No.	Item	Configuration File	Descriptions	Notes
174	Station Position File	file-staposfile	Input the station position file path to retrieve the position from the station list. The station position file is a text file which contains the multiple lines. Each line represents a record for a station. A record contains: -Latitude (deg) -Longitude (deg) -Ellipsoidal height (m) -Station ID -Station name separated by spaces. The line starting ″%″ is treated as the comment line. SINEX station positions can be used as well.	
175	Geoid Data File	file-geoidfile	Input the file path of the geoid data file if selecting the external model as Geoid Model.	
176	IONEX-TEC Data File	file-ionofile	Input the file path of the ionex tec grid data file if selecting the Ionosphere Correction as ionex-tec.	
177	DCB Data File	file-dcbfile	Input the file path of DCB correction for PPP in CODE format.	
178	EOP Data File	file-eopfile	Input the file path of an EOP data file. The format of the EOP data file shall be IGS ERP format version.2. Reference : [IGSMAIL-1943] AC Coordinator, New IGS ERP Format (version 2), July 10, 1998	
179	Ocean Loading BLQ Format	file-blqfile	Input the file path of an OTL coefficients file. The format of the OTL coefficients file is BLQ format. Reference : Ocean Tide Loading Provider (http://holt.oso.chalmers.se/loading)	
180	File Path of fcb Product	file-fcbfile	Input the file path of the fcb product file.	
181	File Path of BIA Product	file-biafile	Input the file path of the BIA Product(Bias SINEX) file.	
182	FTP/HTTP Local Directory	file-tempdir	Input the local directory for FTP/HTTP download. The downloaded files are save in the directory.	RTKRCV only
183	File Path of Stat	file-solstatfile	Onput the file path of an stat data file.	
184	File Path of Trace	file-tracefile	Onput the file path of an trace file.	
185	Azimuth/Elevation Mask Definition File	file-maskfile	Specifies the path to the azimuth/elevation mask definition file.	

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No.	Item	Configuration File	Descriptions	Notes
186	Satellite Exclusion Configuration File	file-exclsatsfile	Specifies the path to the satellite exclusion configuration file.	
187	PPP Kalman Filter Output File	file-pppkf	Specify the file path for the PPP Kalman filter state vector and covariance output.[*.kf] * Will be replaced with the actual output file name.	

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Table 7 RTKRCV specific configuration file format

No.	Item	Configuration File	Descriptions	Notes
1	Password	console-passwd	Login password for remote console ("": no password)	
2	Time Type	console-timetype	Define the time format for displaying the solution in the console. (0:gpst,1:utc,2:jst,3:tow)	
3	Solution Type	console-soltype	Define the solution format for displaying the solution in the console. (0:dms,1:deg,2:xyz,3:enu,4:pyl)	
4	Solution Flag	console-solflag	Define the solution flag for displaying the solution in the console. (0:off,1:std+2:age/ratio/ns)	
5	Input Stream Type	inpstr1-type inpstr2-type inpstr3-type inpstr4-type	Define the input stream type. (inpstr1 to inpstr4) (0:off,1:serial,2:file,3:tcpsvr,4:tcpcli,6:ntripcli,7:ftp,8:http,10:udpsvr,11:udpcli) -OFF : No input stream is used. -SERIAL : Input is received via a serial port. -FILE : Input is read from a file. -TCPSVR : Input is received from a TCP server. -TCPCLI : Input is received from a TCP client. -NTRIPCLI : Input is received from an NTRIP client (typically used for GNSS correction data). -FTP : Input is received via FTP. -HTTP : Input is received via HTTP. -UDPSVR : Input is received from a UDP server. -UDPCLI : Input is received from a UDP client.	
6	Input Stream Path	inpstr1-path inpstr2-path inpstr3-path inpstr4-path	Define the input stream path for stream. (inpstr1 to inpstr4) The format depends on the selected stream type. Stream path formats: - SERIAL : port[:bit_rate[:byte[:parity(n|o|e)[:stopb[:fctr(off|on)]]]]][#port*1] - FILE : path[::T[::+offset][::xspeed]] - TCPSVR : :port - TCPCLI : addr:port - NTRIPCLI : [user[:passwd]]@addr:port/mntpnt - FTP : user:passwd@addr/path[::T=poff,tint,off,rint] - HTTP : addr/path[::T=poff,tint,off,rint] - UDPSVR : :port - UDPCLI : addr:port	

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No.	Item	Configuration File	Descriptions	Notes
7	Input Stream Format	inpstr1-format inpstr2-format inpstr3-format inpstr4-format	Define the data format of the input stream (inpstr1 to inpstr4). This setting determines how GNSS data is interpreted. (0:rtcm2,1:rtcm3,2:oem4,3:oem3,4:ubx,5:ss2,6:hemis,7:skytraq,8:javad,9:nvs,10:binex,11:rt17,12:sbf,14:sp3,16:l6) - RTCM2 : RTCM Version 2 format (GNSS correction data) - RTCM3 : RTCM Version 3 format (GNSS correction data) - OEM4 : NovAtel OEM4 binary format - OEM3 : NovAtel OEM3 binary format - UBX : u-blox proprietary binary format - SS2 : NovAtel SuperStar II format - HEMIS : Hemisphere GNSS binary format - SKYTRAQ : SkyTraq binary format - JAVAD : JAVAD GNSS binary format - NVS : NVS binary format - BINEX : BINEX (Binary Exchange Format for GNSS data) - RT17 : Trimble RT17 format - SBF : Septentrio Binary Format - SP3 : Standard Product 3 format (precise satellite ephemeris) - L6 : L6E format	
8	Input Stream NMEA Request	inpstr2-nmeareq	Defines the type of NMEA request sent to the base or NRTK input stream. (0:off,1:latlon,2:single) -OFF : Send no request -LATLON : Send a request to transmit the configured base station latitude and longitude -SINGLE : Send a request for a single positioning solution	
9	Input Stream NMEA Latitude	inpstr2-nmealat	Set the latitude of the base station to be transmitted.	
10	Input Stream NMEA Longitude	inpstr2-nmealon	Set the longitude of the base station to be transmitted.	

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No.	Item	Configuration File	Descriptions	Notes
11	Input Stream SSR	inpstr1-ssr inpstr2-ssr inpstr3-ssr inpstr4-ssr	Defines the type of SSR (State Space Representation) data input for each stream (inpstr1 to inpstr4). This setting specifies how SSR data is interpreted depending on the source. -MAIN : Receive SSR data from the main antenna -SUB : Receive SSR data from the sub antenna -OFF: SSR input is disabled Note: Each of MAIN and SUB can be assigned only once across all streams (inpstr1 to inpstr4). For example, MAIN may be assigned to inpstr1, SUB to inpstr2 — but MAIN cannot be assigned to multiple streams.	
12	Output Stream Type	outstr1-type outstr2-type	Define the output stream type. (outstr1 to outstr2) (0:off, 1:serial, 2:file, 3:tcpsvr, 4:tcpcli, 5:ntripsvr, 9:ntripcas, 10:udpsvr, 11:udpcli) - OFF : No output stream is used. - SERIAL : Output is sent via a serial port. - FILE : Output is written to a file. - TCPSVR : Output is sent to a TCP server. - TCPCLI : Output is sent to a TCP client. - NTRIPSVR : Output is sent to an NTRIP server (used to broadcast GNSS correction data). - NTRIPCAS : Output is sent to an NTRIP caster (used to relay GNSS correction streams). - UDPSVR : Output is sent to a UDP server. - UDPCLI : Output is sent to a UDP client.	

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No.	Item	Configuration File	Descriptions	Notes
13	Output Stream Path	outstr1-path outstr2-path	Define the outpt stream path for stream. (outstr1 to outstr2) The format depends on the selected stream type. Stream path formats: - SERIAL : port[:bit_rate[:byte[:parity(n|o|e)[:stopb[:fctr(off|on)]]]]][#port*1] - FILE : path[::T[::+offset][::xspeed]] - TCPSVR : :port - TCPCLI : addr:port - NTRIPSVR : :passwd@addr:port/mntpnt[:str*2] - NTRIPCAS : [user[:passwd]]@:port/mntpnt[:str*2] - FTP : user:passwd@addr/path[::T=poff,tint,off,rint] - HTTP : addr/path[::T=poff,tint,off,rint] - UDPSVR : :port - UDPCLI : addr:port	
14	Output Stream Format	outstr1-format outstr2-format	Define the data format of the output stream (outstr1 to outstr2). This setting determines how positioning solution data is formatted. (0:llh, 1:xyz, 2:enu, 3:nmea, 4:stat) -LLH : Solution format: latitude/longitude/height (geodetic coordinates) -XYZ : Solution format: x/y/z in ECEF (Earth-Centered, Earth-Fixed) coordinates -ENU : Solution format: east/north/up baseline vector (relative to a reference point) -NMEA : Solution format: NMEA-0183 standard (used by many GNSS receivers) -STAT : Solution format: solution status information (e.g., fix type, number of satellites)	

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No.	Item	Configuration File	Descriptions	Notes
15	Logging Stream Type	logstr1-type logstr2-type logstr3-type logstr4-type	Define the logging stream type (logstr1 to logstr4). This setting determines how solution and observation data are logged. (0:off, 1:serial, 2:file, 3:tcpsvr, 4:tcpcli, 5:ntripsvr, 9:ntripcas, 10:udpsvr, 11:udpcli) -OFF : No logging is performed. -SERIAL : Data is logged from a serial port. -FILE : Data is logged to a file. -TCPSVR : Data is logged from a TCP server. -TCPCLI : Data is logged from a TCP client. -NTRIPSVR : Data is logged to an NTRIP server (used to broadcast GNSS correction data). -NTRIPCAS : Data is logged to an NTRIP caster (used to relay GNSS correction streams). -UDPSVR : Data is logged from a UDP server. -UDPCLI : Data is logged from a UDP client.	
16	Logging Stream Path	logstr1-path logstr2-path logstr3-path logstr4-path	Define the logging stream path. (logstr1 to logstr4) The format depends on the selected stream type. Stream path formats: - SERIAL : port[:bit_rate[:byte[:parity(n|o|e)[:stopb[:fctr(off|on)]]]]][#port*1] - FILE : path[::T[::+offset][::xspeed]] - TCPSVR : :port - TCPCLI : addr:port - NTRIPSVR : :passwd@addr:port/mntpnt[:str*2] - NTRIPCAS : [user[:passwd]]@:port/mntpnt[:str*2] - UDPSVR : :port - UDPCLI : addr:port	
17	Transition Threshold for Dead Reckoning	misc-maxdrtr	Defines the transition threshold for switching to INS dead reckoning. (s)	
18	Processing Cycle/ Buffer Size	misc-svrcycle	Defines the processing cycle time of in ms. Usually set 100 ms or less value. Defines the input message buffer size in bytes. Usually set it to 32768 or more.	
19	Timeout	misc-timeout	Defines the timeout for TCP client and NTRIP client connections in ms. If the timeout time expired without sever response, RTKRCV retries to connect to server after waiting for the re-connect interval.	

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No.	Item	Configuration File	Descriptions	Notes
20	Re-Connect Interval	misc-reconnect	Defines the re-connect interval for TCP client and NTRIP client connections in ms.	
21	NMEA Cycle/ File Swap Margin	misc-nmeacycle	Defines the NMEA GPGGA transmission cycle to NRTK server in ms.	
22	Solution Buffer/Log Size	misc-buffsize	Defines the internal solution buffer size and log size in epochs.	
23	Navigation Message Selection	misc-navmsgsel	Defines the navigation messages to be used. (0:all,1:rover,2:base,3:corr) - All: In any of the input streams - Target: In the target receiver stream - Base Station: In the base station receiver stream - Correction: In the correction stream	
24	HTTP/NTRIP Proxy	misc-proxyaddr	Defines the HTTP/NTRIP proxy server address and port number as <address>:<port> form to connect the NTRIP Caster via a HTTP-Proxy server.	
25	File Swap Overlap Time	misc-fswapmargin	Define the file swap overlap time in seconds. This setting specifies the overlapping period between the previous and new output files to avoid missing transient data during file swapping.	

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Table 8 Signal Identification

SATELLITE SYSTEM	TARGET SIGNAL	TARGET OBSERVATION DATA IDENTIFICATION(RINEX)	XX
GPS	L1C/A	C1C	1C
	L2P	C2P,C2Y,C2W,C2C,C2M,C2N,C2D	2P
	L2C	C2L,C2X,C2S	2L
	L5	C5Q,C5X,C5I	5Q
GLONASS	G1	C1C,C1P	1C
	G2	C2P,C2C	2P
GALILEO	E1	C1C,C1B,C1X	1C
	E5a	C5Q,C5X,C5I	5Q
	E5b	C7Q,C7X,C7I	7Q
	E6	C6C,C6X,C6E	6Q
QZSS	L1C	C1L,C1X,C1S	1L
	L1C/A	C1C	1C
	L1C/B	C1E	1E
	L5	C5Q,C5X,C5I	5Q
	L2C	C2L,C2X,C2S	2L
BEIDOU	B1I	C2Q,C2X,C2I	2Q
	B3I	C6Q,C6X,C6I	6Q
	B2I	C7Q,C7X,C7I	7Q
	B2a	C5P,C5X,C5D	5P

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B.4 Satellite Exclusion Settings File

In the post-processing and real-time positioning software, satellites can be excluded at arbitrary epochs by specifying the path of the satellite exclusion settings file in file-exclsatsfile. The file format is shown in Table 9.

Table 9 Satellite Exclusion Settings File Format

No	Item	Format	Description
1	START EPOCH	YYYY/MM/ DD hh:mm:ss	Enter the date and time in text format as Year/Month/Day Hour:Minute:Second. The time system is GPST. If the field is left blank, the exclusion period is interpreted as lasting until the END EPOCH in the same row. If the END EPOCH in the same row is also left blank, the corresponding entry becomes invalid.
2	END EPOCH	YYYY/MM/ DD hh:mm:ss	Enter the date and time in text format as Year/Month/Day Hour:Minute:Second. The time system is GPST. If the field is left blank, the exclusion period is interpreted as starting from the START EPOCH in the same row. If the START EPOCH in the same row is also left blank, the corresponding entry becomes invalid.
3	SAT	(Refer to the right)	As with pos1-exclsats, specify the satellites to be excluded in RTKLIB format, separated by commas. (Example: G03, J07, etc.)

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B.5 Azimuth and Elevation Mask Definition

In the post-processing and real-time positioning software, specify the path to the azimuth/elevation mask definition file using file-maskfile.

The file format is shown in Table 10. Lines whose first character is '#' are excluded from processing.

Table 10 Azimuth and Elevation Mask Definition format

No	Item	Description
1	Latitude	Express the latitude as a real number from -90 to 90 degrees.
2	Longitude	Express the longitude as a real number from −180 to 180 degrees.
3	Height	Express the ellipsoidal height (WGS84) as a real number.(m)
4	Object	Specify using one character selected from the list below. S : Structure T : Terrain V : Vegetation X : eXtra(used for user-defined custom settings)
5	Mask Type	Specify using one character selected from the list below. H : Horizontal mask definition When Gn is set to 45, the mask covers the range from 0° to 45° measured from the horizontal direction. Z : Zenith mask definition When Gn is set to 45, the mask covers the range from 45° to 90° measured from the zenith direction.
6	AZ interval	Specify the azimuth definition interval as a real number.(deg) The azimuth (AZ) is defined as 0° at north, 90° at east, 180° at south, and 270° at west.
7	Mask EL	Specify the elevation mask (1 to 360/F) as a real number. The elevation (EL) is defined as 0° at the horizontal and 90° at the zenith.

sample:

```
#Lat.,Lon.,Height,Object,Type,AZ int.,0, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330
36.081304679, 140.113717014, 69.795, T, H, 30, 7, 16, 25, 51, 65, 0, 5, 20, 11, 46, 13, 0
36.081304679, 140.113717014, 69.795, T, Z, 30, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90
36.081304679, 140.113717014, 69.795, V, H, 30, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
36.081304679, 140.113717014, 69.795, V, Z, 30, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90
```


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B.6 Output of states and covariance

In the post-processing and real-time positioning software, state and covariance outputs are generated by defining -opt and file -pppkf. Lines beginning with '#' are treated as comment lines.A data-item description line is placed before the start of the data section.

The output items vary depending on the state variables, the covariance output level, and the positioning conditions.The file format is shown in Table 11.

Table 11 Output of states and covariance format

No.	Item	Descriptions	Unit	Note
1	GPST	GPS epoch time	-	YYYY/MM/DD hh:mm:ss.000
2	Q	quality status	-	The source of the state outputs varies depending on the value of Q. Q=1:rtk->xa Q=6:rtk->x
3	POS-X, POS-Y, POS-Z	position{x,y,z}	m	Output when pos1-dynamics is off or set to dynamics.
4	VEL-X, VEL-Y, VEL-Z	velocity{x,y,z}	m/s	Output when pos1-dynamics is dynamics.
5	ACC-X, ACC-Y, ACC-Z	acceleration{x,y,z}	m/s^2	Output when pos1-dynamics is dynamics.
6	dPOS-X, dPOS-Y, dPOS-Z	delta position{x,y,z}	m	Output when pos1-dynamics is ins.
7	dVEL-X, dVEL-Y, dVEL-Z	delta velocity{x,y,z}	m/s	Output when pos1-dynamics is ins.
8	dATT-R, dATT-P, dATT-H	delta attitude {roll,pitch,heading}	deg	Output when pos1-dynamics is ins.
9	dACCB-X, dACCB-Y, dACCB-Z	delta accelerometer bias{x,y,z}	m/s^2	Output when pos1-dynamics is ins.
10	dGYRB-R, dGYRB-P, dGYRB-H	delta gyro bias{roll,pitch,heading}	deg/s	Output when pos1-dynamics is ins.
11	CLKD	clock drift	ns	Output when pos1-dynamics is ins.
12	CLKGPS	clock bias(GPS)	ns	Satellite systems excluded or not observed in pos1-navsys are not estimated, but output is still produced for all allocated systems.
13	CLKGLO	clock bias(GLO)	ns	-
14	CLKGAL	clock bias(GAL)	ns	-
15	CLKQZS	clock bias(QZS)	ns	-
16	CLKBDS	clock bias(BDS)	ns	-
17	CLKIRN	clock bias(IRN)	ns	-
18	TRP	tropospheric delay	m	Output when pos1-tropopt is est-ztd or set to est-ztdgrad.

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No.	Item	Descriptions	Unit	Note
19	GRAD-N	tropospheric delay gradient(north)	-	Output when pos1-tropopt is est-ztdgrad.
20	GRAD-E	tropospheric delay gradient(east)	-	Output when pos1-tropopt is est-ztdgrad.
21	ION-{X}	vertical ionospheric delay X：satellite ID(G01～)	m	Output when pos1-tropopt is est-stec. Unobservable satellites are not estimated, but all allocated satellites are still output.
22	BDSB-{X}	inter system bias X：number of satellite type(1～3)	ns	Output when BeiDou is included in pos1-navsys.
23	AMB{X}-{Y}	ambiguity X：number of freqency type Y：satellite ID(G01～)	m	Unobservable satellites are not estimated, but all allocated satellites are still output.
24	c{X}	diagonal covariance X：name of states item	-	Output when the level is 2 or higher. Only the diagonal elements of the covariance are output.
25	c{X}	full covariance X：name of states item	-	When level is 3, only the final epoch is output. Because the data array differs, the data item description line (#) is output first, followed by the data. The data is formatted as a 2-dimensional representation, with each row corresponding to the number of state variables.

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Appendix C : Details of newly added functions

C.1 PPP-AR function

MALIB has modified the PPP-AR function source file “ppp_ar.c” included in RTKLIB 2.4.3 b34 to add the ability to solve AR in PPP as well. In addition, in the MALIB, GPS, GAL, and QZS, excluding GLONASS which are FDMA signals, can be input to AR processing. As shown in Table 6, “pos2-arsys” in the configuration file allows GNSS can be selected. When using Cascading Ambiguity Resolution, it can be enabled by setting the “pos2-car” parameter to 1 (on).

C.2 Input function compliant with L6E data format for each receiver

MALIB support L6E proprietary formats (e.g. .sbf, .ubx,) of each receiver in addition to RTCM3 format to perform MADOCA-PPP. A process to read L6E correction information in Compact-SSR format was added to MALIB by reusing the existing correction data process related to the RTCM3 SSR format. Figure 1 and Figure 2 shows a flowchart of the data and processing flow in post-processing and real-time processing.

Table 12 Combination of satellite systems that solve AR,
using bit array similar to pos1-navsys

	NAV	BDS	QZS	GAL	GLO	SBAS	GPS	VALUE (DEC)
GPS	0	0	0	0	0	0	1	**1**
GAL	0	0	0	1	0	0	0	**8**
GPS+GAL	0	0	0	1	0	0	1	**9**
QZS	0	0	1	0	0	0	0	**16**
GPS+QZS	0	0	1	0	0	0	1	**17**
GAL+QZS	0	0	1	1	0	0	0	**24**
GPS+GAL+QZS	0	0	1	1	0	0	1	**25**

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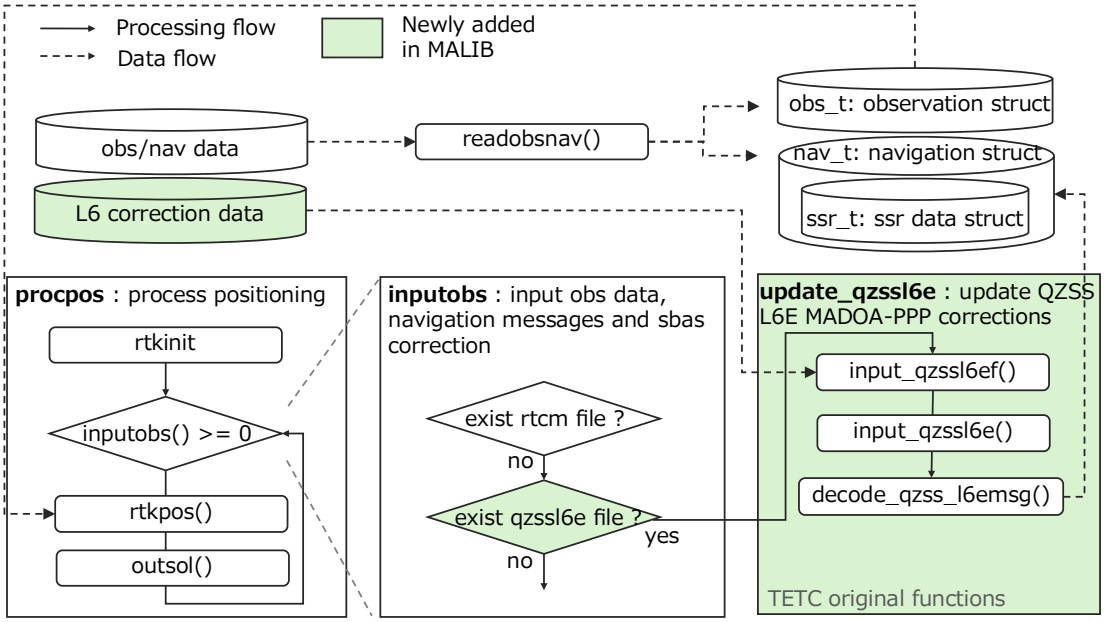


Figure 1 L6E correction data processing flow (post-processing)

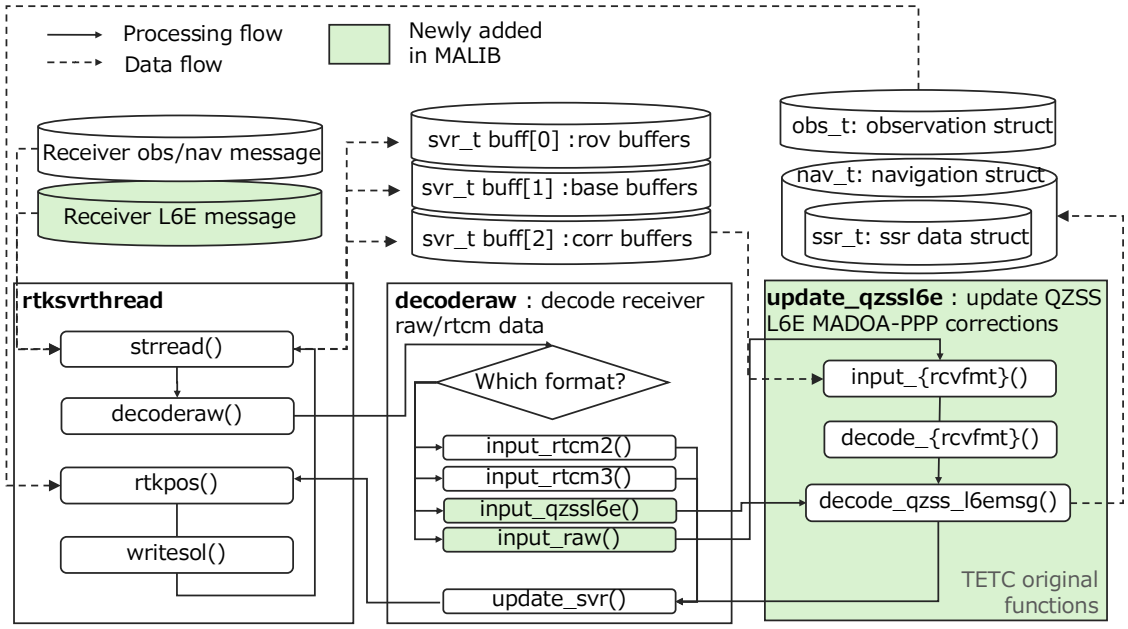


Figure 2 L6E correction data processing flow (real-time-processing)

The epoch time of the MADOCA-PPP correction information is defined as “GPS epoch time” or “GNSS epoch time” and no week number is defined. In the case of post-processing, the epoch time of the observation data is used to calculate the correction week number. On the other hand, in the case of real-time processing, the current time of the computer is used as the basis for processing, which may

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cause problems when replaying past data to check operation. Therefore, MALIB uses "-rst" command as a startup option and has a function to specify the playback start time. (*Note: The computer time must be adjusted to the time of the past data.)

Table 13 shows the L6E format of each receiver supported by MALIB: U-blox and Septentrio receivers store one QZSS L6E data per epoch, and MSJ receivers store three QZSS L6E data per epoch. When decoding data with multiple QZSS PRNs per epoch, such as MSJ receivers, the first QZSS PRN processed is continued to be used as the processing target. If the data in the QZSS PRN to be processed is interrupted due to satellite invisibility, etc., MALIB switches to another QZSS PRN to be processed.

Table 13 Each L6E receiver format supported in MALIB

Class&ID	***Message***	***Content description***	***L6E file extension***
0x0273	UBX-RXM-QZSSL6	QZSS L6 message	.ubx (Ublox)
4069	QZSSRawL6	QZSS L6 navigation message	.sbf (Septentrio)
4070	QZSSRawL6D	QZSS L6D navigation message	.sbf (Septentrio)
4071	QZSSRawL6E	QZSS L6E navigation message	.sbf (Septentrio)
4050		QZSS LEX signal raw data	.rtcm3 (Magellan Systems Japan (MSJ))

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C.3 Signal selection function

The combination of GNSS satellite signals used can be switched by settings (GPS_IIR-M, GPS_IIF, GPSIIIA, QZSS1-2, QZSS-2A, QZSS-3, GAL, BDS-2, BDS-3) as shown in Table 14. For satellites other than those defined here, L1C/A-L2P for GPS, L1C-L5 for QZSS, E1-E5a for Galileo, and G1-G2 for GLONASS. When performing PPP, the "pos1-frequency" option in the configuration file must be changed according to the frequency combination.(ex. For three frequencies, set pos1-frequency to "l1+2+3".)

Table 14 Signal type and frequency setting in MALIB

GNSS	Signal Type	pppobs[0]	pppobs[1]	pppobs[2]	pppobs[3]	Setting
GPS	IIR	L1C/A	L2P	-	-	-
	IIRm	L1C/A	L2P	-	-	0
			L2C	-	-	1
			L2*		-	3
	IIF	L1C/A	L2P	-	-	0
			L2C	-	-	1
			L5	-	-	2
			L2*		-	3
			L2P	L5	-	4
			L2C	L5	-	5
	IIIa	L1C/A	L2P	-	-	0
			L2C	-	-	1
			L5	-	-	2
			L2*	-	-	3
			L2P	L5	-	4
			L2C	L5	-	5
QZSS	Q2-Q4	L1C	L5	-	-	0
		L1C/A	L2C	-	-	1
		L1C	L2C	L5	-	3
		L1C/A	L2C	L5	-	4
	Up to 1R	L1C	L5	-	-	0
		L1C/A	L2C	-	-	1
		L1C/B	L5	-	-	2
		L1C	L2C	L5	-	3
		L1C/A	L2C	L5	-	4
		L1C/B	L2C	L5	-	5
	Q5-Q7	L1C	L5	-	-	0
		L1C/B	L5	-	-	2
Galileo		E1	E5	-	-	0
			E5b	-	-	1
			E5a	E5b	-	2
			E5a	E5b	E6	3

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GNSS	Signal Type	pppobs[0]	pppobs[1]	pppobs[2]	pppobs[3]	Setting
GLONASS		G1	G2	-	-	-
BeiDou	BDS2	B1I	B3I	-	-	0
			B2I	-	-	1
			B3I	B2I	-	2
	BDS3	B1I	B3I	-	-	0
			B2a	-	-	1
			B3I	B2a	-	2

* Either the L2C or L2P signal is utilized.

C.4 PPP/PPP-AR with single-station local area atmospheric corrections

MALIB supports local atmospheric corrections and can shorten the initial convergence time of PPP/PPP-AR. It can input atmospheric corrections in RTKLIB STAT format for both post-processing and real-time processing. The $ION line is processed as ionospheric correction, and the $TROP line is processed as tropospheric correction. The $POS line is also used as the generation position for correction information. Figure 3 and Figure 4 show the data I/O in post-processing and real-time-processing.

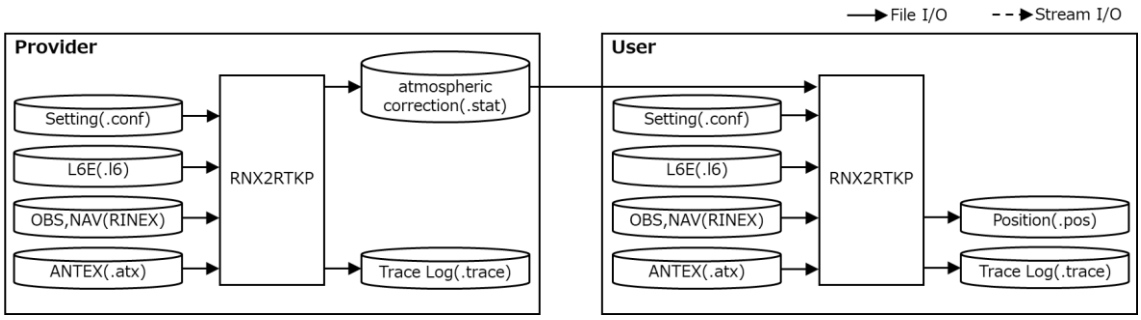


Figure 3 Single-station atmospheric correction flow (post-processing)

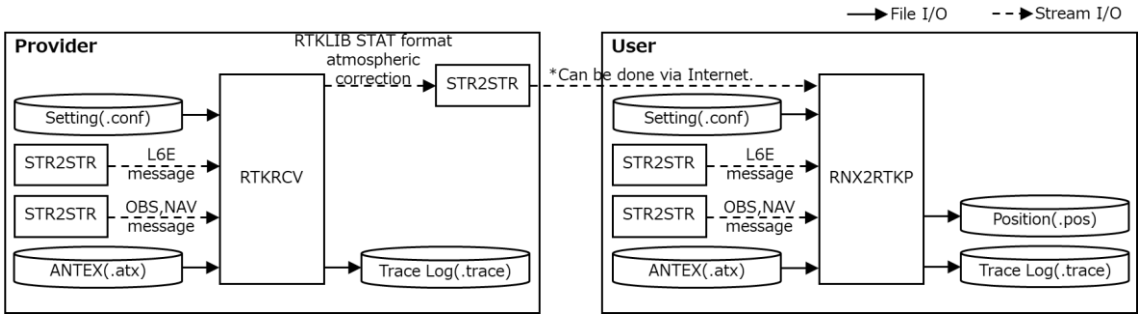


Figure 4 Single-station atmospheric correction flow (real-time-processing)

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C.5 Substitute positioning mode

MALIB provides two alternative navigation techniques to ensure accurate positioning even when QZSS L6E signals are unavailable or ground-based communication is interrupted. These techniques are categorized based on their dependency on ground communication and the source of ephemeris and clock data. The substitute positioning mode can be selected by setting the pos1-subopt option in the configuration file.

(1) PPP using broadcast ephemeris from GNSS satellites

This method uses only broadcast navigation messages from GNSS satellites, without relying on external precise products. It requires no ground communication and is suitable for fully autonomous onboard operation. However, positioning accuracy and convergence speed are limited.

(2) PPP using real-time precise ephemeris from JAXA via internet link

This method uses real-time orbit and clock corrections provided by JAXA and delivered via inter-satellite communication or other space-based data links. It offers high positioning accuracy but requires continuous ground-based data transmission.

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C.6 Dead reckoning

MALIB uses dead reckoning to maintain positioning continuity during GNSS signal outages.

(1) Post-processing

When the temporal gap between successive GNSS observations exceeds the predefined Dead Reckoning Interval (pos1-drint), dead reckoning is employed by interpolating the trajectory at fixed intervals, as specified by the corresponding parameter.

An example of interpolation for dead reckoning in post-processing is shown in Figure 5.

$$\text{Number of interpolations} = \text{trunc}((t_i - t_{i-1})/DR_{int} + 0.5) - 1$$
$$= \text{trunc}((7-0)/1.0+0.5)-1$$
$$= 6$$

t_i :Time of current observation
t_{i-1} :Time of previous observation
DR_{int} :Dead Reckoning Interval (pos1-drint)

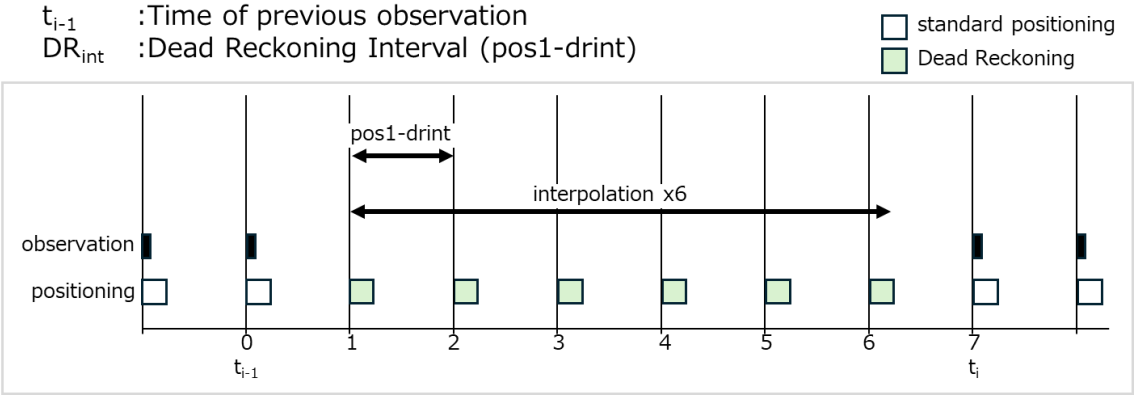


Figure 5 Timing of dead reckoning (post-processing)

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(2) Real-time-processing

In the process of positioning result output, the system retains the computer timestamp and continuously monitors the elapsed time during the waiting period for incoming observational data. When the elapsed time surpasses the threshold defined by a user-specified parameter, dead reckoning is initiated to compensate for the absence of real-time measurements.

The timing of dead reckoning in real-time-processing is shown in Figure 6.

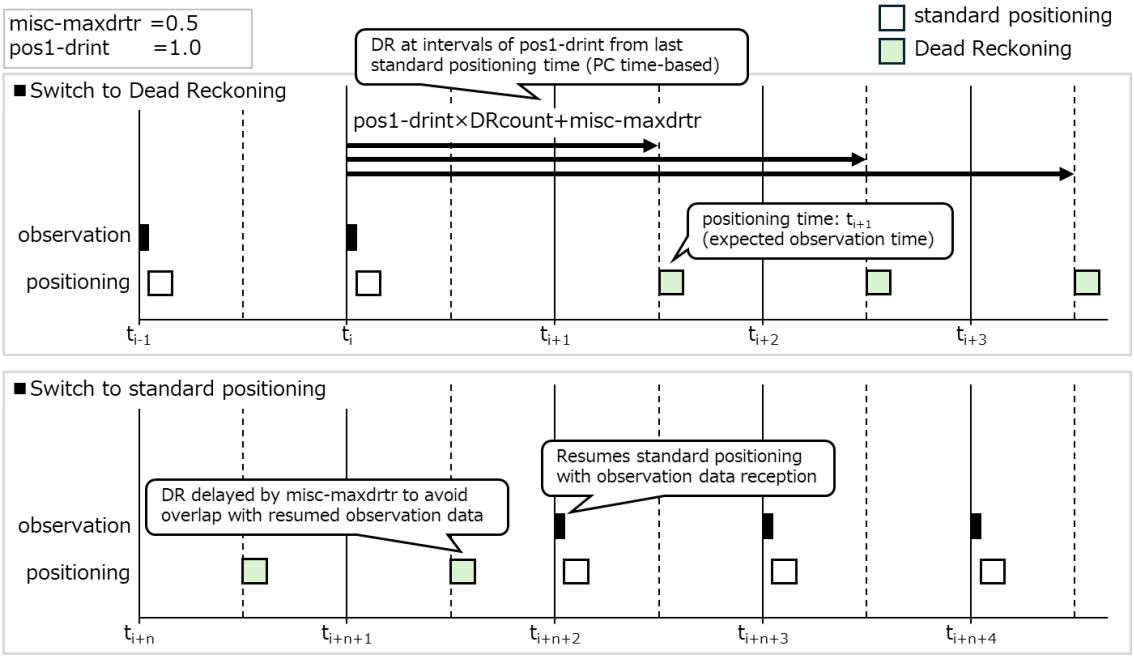


Figure 6 Timing of dead reckoning (real-time-processing)

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C.7 Code and phase bias settings

Satellite code bias settings

The method for applying satellite code bias correction is configured using Satellite Code Bias Option for PPP (pos2-pppsatcb), which is set in the Configuration File.

0: auto – Recommended for general purposes. Automatically selects the first available valid bias correction method in the order: L6E, BIA, DCB, RTCM3

Other available options:

- 1: l6e – Uses L6E signal-based bias correction
- 2: bia – Uses satellite code bias data obtained from Bias SINEX
- 3: dcb – Uses Differential Code Bias data
- 4: rtcm3 – Applies RTCM Version 3 standard bias correction

Notes

- BIA applies OSB values obtained from the Bias SINEX file (.bia) specified by file-biafile.
- DCB applies Differential Code Bias values obtained from the DCB file (.dcb) specified by file-dcbfile.

When using JAXA-MADOCA ephemeris, it is recommended to apply satellite code bias using either BIA or DCB methods.

In the case of real-time ephemeris, satellite code bias is updated every 3 hours.

To define the validity period of satellite code bias correction, specify the following in misc-rtcmopt. The value represents the validity duration in seconds:

-RTCM_CB_VALID=10860

Satellite phase bias settings

Satellite phase bias values are applied from either L6E signal or real-time ephemeris products.

When using JAXA-MADOCA real-time ephemeris, satellite phase bias is updated approximately every 15 minutes.

To define the validity period of satellite phase bias correction, specify the following in misc-rtcmopt. The value represents the validity duration in seconds:

-RTCM_PB_VALID=960

Recommended configuration

pos2-pppsatcb = auto

misc-rtcmopt = -RTCM_CB_VALID=10860, -RTCM_PB_VALID=960

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